ST. JOSEPH'S COLLEGE OF ENGINEERING

(An Autonomous Institution) • OMR, Chennai - 119

LAB RECORD

Professional English Lab Integrated

CLASS CODE: EN25C09

STUDENT DETAILS

Student Name: _____

Roll No: _____

Date: _____

Subject Name: _____

DEPARTMENT OF HUMANITIES & SCIENCES • AFFILIATED TO ANNA UNIVERSITY • CHENNAI

Self Introduction: Written Script and One-Page Profile

IRFANA S | First-year B.E. Computer Science and Engineering

Self Introduction – Written & One Page Profile

IRFANA S

Aspiring Machine Learning Engineer | Generative AI

Pudukkottai, Tamil Nadu, India | B.E. Computer Science & Engineering Student
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PROFESSIONAL SUMMARY

First-year B.E. Computer Science and Engineering student with a strong interest in Artificial Intelligence and Machine Learning, focused on building a solid foundation in intelligent technologies. Currently strengthening programming fundamentals and learning how machine learning algorithms, data-driven techniques and AI models can be applied to solve real-world problems. Areas of interest include predictive modelling, intelligent application development and Generative AI, with a focus on how these technologies improve efficiency and support smarter decision-making. Continuously working to expand technical knowledge, improve problem-solving abilities and gain practical experience through hands-on learning and future projects. Long-term goal: to become a skilled Machine Learning Engineer, contribute to innovative AI/ML solutions and develop reliable, intelligent applications that create meaningful value in the real world.

EDUCATION

Bachelor of Engineering (B.E.) – Computer Science and Engineering
St. Joseph's College of Engineering, Chennai | First Year | Joined 2026
Higher Secondary Education – Bio-Maths: studied Biology and Mathematics.

TECHNICAL SKILL SET

- **Programming:** C Programming – Beginner Level
- **Computer Science:** Programming Fundamentals, Logical Problem-Solving
- **Currently Learning:** C programming fundamentals, logical thinking, problem-solving
- **Preparing to Learn:** Python, data analysis, statistics and machine learning concepts
- **Languages:** English, Tamil

PROFESSIONAL FOCUS

- **Primary Specialisation – Machine Learning Engineering:** learning how data-driven models are built, evaluated and used to make predictions.
- **Additional Interest – Generative AI:** exploring AI-powered applications, chatbots and tools that can generate useful content.

PROJECTS & HANDS-ON LEARNING

Current status: beginner and skill development

- No technical project has been completed yet.
- Learning path: practise C programming, learn Python, then build beginner-friendly projects such as a basic prediction system or an AI-powered study assistant (planned, not yet built).
- Project proposal prepared for this lab: **AI-Based Resume Analyzer** (see Experiment No: II-7).

Self Introduction: Written Script and One-Page Profile

IRFANA S | First-year B.E. Computer Science and Engineering

VALUE-ADDED LEARNING & TRAINING

- **C Programming** – Currently Learning
- **Python Programming** – Preparing to Learn
- **Data Analysis, Statistics and Machine Learning** – Building the foundation
- **Generative AI and AI Application Development** – Exploring

PROFESSIONAL & INTERPERSONAL STRENGTHS

- **Strong Interest in AI:** genuine curiosity about AI and machine learning, which helps me keep going through difficult topics.
- **Clear Direction:** I have chosen my domain (AI and ML), my specialisation (Machine Learning Engineering) and an additional interest (Generative AI) from the first year.
- **Logical Problem-Solving:** developing analytical thinking through programming practice.
- **Continuous Learning:** committed to learning step by step and building a strong technical foundation.
- **Time Advantage:** four full years of study ahead to build skills, projects and experience.

SELF-INTRODUCTION SCRIPT

“Good morning. My name is Irfana S. I am a first-year B.E. Computer Science and Engineering student at St. Joseph's College of Engineering, Chennai, and I completed my higher secondary education in the Bio-Maths stream. My career goal is to become a Machine Learning Engineer, and I am also interested in Generative AI, chatbots and intelligent applications. At present I am learning C programming and strengthening my logical thinking and problem-solving skills, and I am preparing to learn Python, data analysis, statistics and machine learning. I want to build practical projects and contribute to AI/ML solutions that create real-world value. Thank you.”

CAREER DIRECTION

Programming Fundamentals → Python & Machine Learning → Machine Learning Engineering → Generative AI Applications

Impressive Personality: Biographical Note

Andrew Ng

ANDREW NG

Computer Scientist | AI Educator | Entrepreneur and Investor

A biographical note on one of the people who made machine learning learnable worldwide

Born	18 April 1976, United Kingdom, to parents who emigrated from Hong Kong
Education	B.Sc. Carnegie Mellon University; M.Sc. Massachusetts Institute of Technology; Ph.D. University of California, Berkeley (2002)
Known for	Deep learning, Google Brain, Coursera, DeepLearning.AI, AI Fund, democratising AI education
Today	Founder of DeepLearning.AI; Managing General Partner, AI Fund; Executive Chairman, LandingAI; Chairman and co-founder, Coursera; Adjunct Professor, Stanford; Amazon board director

Introduction

Andrew Yan-Tak Ng is a British-born American computer scientist and engineer whose work has shaped both how machine learning is built and how the world learns it. He is known for pioneering research in deep learning, machine perception, computer vision and natural language processing. Just as important, he is known for turning a graduate-level subject into something a curious beginner anywhere can study. His online courses are estimated to have reached around eight million learners.

His career moves through four connected worlds: academic research at Stanford, industrial research at Google and Baidu, online education through Coursera and DeepLearning.AI, and company building through LandingAI and AI Fund. The thread running through all of them is one belief: AI should be practical, widely understood and useful to ordinary businesses and people, not confined to a few research labs.

Early Life and Education

Ng was born in the United Kingdom in 1976 to parents who had immigrated from Hong Kong. He studied at three of the world's leading technical universities: an undergraduate degree at Carnegie Mellon, a master's at MIT, and a doctorate at the University of California, Berkeley, completed in 2002. In 1996 he received the Andrew Carnegie Society Scholarship, and in 2007 he was awarded a Sloan Fellowship. That path, combining computer science, electrical engineering and statistical learning, is the foundation of his later work on systems that learn from data.

Career Timeline

2002	Joins Stanford University as a faculty member in Computer Science and Electrical Engineering; later becomes Director of the Stanford AI Lab (SAIL).
2008	Starts Stanford Engineering Everywhere, publishing Stanford courses online for free, including his Machine Learning course (CS229) lectures.
2011	His applied machine learning class opens online (ml-class.org) with over 100,000 students enrolled. He then co-founds Coursera with Daphne Koller.
Early 2010s	Founds and leads the Google Brain team, an early large-scale deep learning effort inside Google.
2014-2017	Serves as Chief Scientist and Vice President at Baidu, leading a major AI research organisation.
2017	Launches DeepLearning.AI and its deep learning courses; founds LandingAI, bringing AI to manufacturing.
2018	Announces AI Fund with $175 million to build and back new AI startups.
2024	Joins the Amazon board of directors and becomes Executive Chairman of LandingAI.

Compiled October 2026 from public sources. Roles can change; verify before citing. Page 1

Impressive Personality: Biographical Note

Andrew Ng

ANDREW NG | Biographical Note

Major Contributions

Making machine learning teachable. Ng's free Stanford lectures and the 2011 online class showed that a rigorous machine learning course could attract a global audience. That experiment helped launch the modern MOOC movement and led directly to Coursera, which he co-founded and still chairs. In 2019 he added *AI for Everyone*, a non-technical course explaining what AI can and cannot do for business and society.

Large-scale deep learning in industry. At Google Brain and later at Baidu, he helped show that deep neural networks improve dramatically with more data and computing power, and that this could be turned into real products. He has authored or co-authored more than 200 papers in AI and related fields.

Building the AI ecosystem. DeepLearning.AI trains practitioners, including short courses on generative AI and large language models. LandingAI applies computer vision to industrial settings. AI Fund builds companies from the ground up, and he is also a co-founder of AI Aspire.

His Approach to Machine Learning and Generative AI

- **Learn by building.** He consistently pairs theory with hands-on projects, so students ship working systems rather than only study equations.
- **Data and deployment matter.** He has long argued that real value comes from good data and sound engineering around a model, not only from the model itself.
- **AI is a general-purpose tool.** He often compares AI to electricity: a technology that will touch every industry, so everyone should understand it, not just specialists.
- **Responsible, practical adoption.** His current focus is AI training and entrepreneurial ventures that help the global economy adopt AI responsibly, including generative AI.

Recognition

- Named to *Time* magazine's 100 Most Influential People (2012) and to the Time100 AI list (2023).
- Named to Fast Company's Most Creative People list (2014).
- Sloan Fellowship (2007); Andrew Carnegie Society Scholarship (1996).

What a First-Year Student Can Learn from His Path

- **Master the fundamentals first.** His career rests on solid mathematics, programming and statistics, the same base a first-year ECE student is building now.
- **Teach what you learn.** Explaining ideas clearly deepened his own understanding and multiplied his impact.
- **Build projects early.** Start with small models in Python, publish them on GitHub, and grow from classifiers to deployed systems.
- **Stay curious across fields.** He moved between research, education and business without losing the core technical skill.
- **Think about people.** The best AI work solves a real problem for real users.

Research Highlights

Before he became best known as a teacher, Ng was an active researcher in machine learning and robotics. Three threads of that research are worth knowing, because each foreshadows the practical, systems-minded style of his later work.

- **Reinforcement learning and robotics.** At Stanford he worked on teaching robots through learning rather than hand-written rules, including work on autonomous helicopter flight. The lesson: learning methods can handle real-world control problems that are hard to program by hand.
- **Large-scale deep learning at Google.** The Google Brain project, a collaboration with Jeff Dean and others, showed that very large neural networks trained on huge amounts of unlabeled data could learn useful features on their own. This helped convince industry that scale was a key ingredient of deep learning.
- **Unsupervised and representation learning.** His work on learning useful representations from raw data shaped modern deep networks and relates to how generative models are trained.

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Impressive Personality: Biographical Note

Andrew Ng

ANDREW NG | Biographical Note

Ventures and Organisations

Organisation	Role	What it does
Coursera	Co-founder, Chairman	Online learning platform offering courses and degrees from universities and companies worldwide.
DeepLearning.AI	Founder	AI education: specialisations, short courses and the weekly newsletter *The Batch*.
LandingAI	Founder, Executive Chairman	Applies computer vision and AI to manufacturing and other industrial use cases.
AI Fund	Managing General Partner	Venture studio that builds and funds AI startups from the ground up.
Stanford University	Adjunct Professor	Computer Science; formerly Director of the Stanford AI Lab.
Amazon	Board Director	Joined the board in April 2024.

Courses and Resources Associated with Ng

These are good starting points if you want to follow his teaching directly. Check the course pages for current content and availability, since offerings are updated over time.

- **Machine Learning (Stanford / Coursera).** The course that introduced a generation to supervised learning, regression, neural networks and practical advice on applying them.
- **Deep Learning Specialization.** A sequence on neural networks, sequence models and structuring machine learning projects.
- **AI for Everyone.** A non-technical overview of AI strategy and its effect on business and society.
- **Generative AI and LLM short courses.** Short, hands-on courses on topics such as prompt engineering, building with large language models, and retrieval-augmented applications.
- **The Batch.** A weekly newsletter on AI news and trends, written for practitioners.

A Suggested Roadmap for Your Own Journey

A practical way for a first-year student to follow the same ideas, step by step. This is a suggestion, not a rule; adjust the pace to your workload.

Year 1	Strengthen Python and C, linear algebra, calculus and basic probability. Build small projects and keep them on GitHub.
Year 2	Learn core machine learning: regression, classification, model evaluation. Practice with NumPy, Pandas and scikit-learn on public datasets.
Year 3	Study deep learning with PyTorch or TensorFlow. Try computer vision and NLP projects, then explore generative AI and LLM applications.
Year 4	Build an end-to-end project: data, training, deployment and monitoring. Prepare for internships and ML engineering roles.

> *Takeaway: Andrew Ng's story shows that a career in machine learning engineering and generative AI begins with strong basics, steady practice and the willingness to share what you know.*

Sources: Wikipedia (Andrew Ng); World Economic Forum speaker profile; Interesting Engineering biography; Amazon director profile (VectorShift); MarketScreener insider profile; DeepAI profile.

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Summary of Lecture Notes

NPTEL Soft Skills course – weekly learning summaries (Weeks 1–8)

NPTEL | IIT KANPUR | 8-WEEK COURSE SUMMARY

Enhancing Soft Skills and Personality

Course instructor: Prof. T. Ravichandran, Department of Humanities and Social Sciences

Course code 109104115 | A week-by-week study summary

Course	NOC: Enhancing Soft Skills and Personality (NPTEL, IIT Kanpur)
Instructor	Prof. T. Ravichandran, Professor of English, Department of Humanities and Social Sciences, IIT Kanpur
Format	8 weeks, with about five short lectures each week
Audience	Students, teachers, professionals, trainers, leaders and employers; approved by AICTE for Faculty Development Programmes

Course Overview

This course is about the everyday skills that decide how far technical knowledge actually takes a person: how you think about your own ability, how you use your time, how you deal with other people, how you speak and write, and how you look after your body, mind and money. It builds on Prof. Ravichandran's earlier NPTEL course on developing soft skills and personality, and revisits that course's highlights in Week 1 before moving on to new topics.

The eight weeks move from the inside out. They begin with mindset, then cover time and self-management, assertiveness and emotions, and how others see you. They then turn to language errors, humour and money, health and relationships, and finally ethics, etiquette, memory and care for the environment.

The Eight Weeks at a Glance

Week	Theme	Core focus
1	Mindset	Growth mindset and learning attitudes
2	Time and procrastination	Using time well, avoiding delay
3	Assertiveness and emotions	Saying no, controlling anger, positive thinking
4	Likeability and appeal	How others see you and how to be attractive
5	Common errors	Avoiding frequent mistakes in communication
6	Humour and money	Workplace humour, managing money
7	Health and love	Exercise, diet, sleep, relationships
8	Ethics, etiquette and mind	Etiquette, memory, environment

Week-by-Week Summary

WEEK 1 Mindset: Foundations and Growth Mindset

Lectures: Highlights of the earlier Developing Soft Skills and Personality course (two parts); Definitions and types of mindset; Learning mindsets; Secrets of developing growth mindsets.

The course opens by recalling the main ideas of the instructor's earlier soft skills course, so every learner starts from the same base. It then focuses on mindset, the set of beliefs a person holds about themselves and the world. Different mindsets lead to different reactions to challenge, criticism and failure.

The central contrast is between a fixed mindset, which treats ability as something you either have or lack, and a growth mindset, which treats ability as something that develops through effort, good strategies and feedback. The week ends with practical ways of building growth mindsets in your own learning.

Key takeaway: How you think about your abilities shapes how you learn, so adopt a growth mindset first.

Summary of Lecture Notes

NPTEL Soft Skills course – weekly learning summaries (Weeks 1–8)

Enhancing Soft Skills and Personality | 8-Week Summary

WEEK 2 Time Management and Procrastination

Lectures: Importance of time and perceptions of time; Using time efficiently; Understanding procrastination; Overcoming procrastination; Don't say "yes" to make others happy.

Week 2 treats time as a limited resource and looks at how different people perceive it. It then covers using time efficiently, which means deciding what matters, planning around it and protecting your attention.

Two lectures are devoted to procrastination: first understanding why people delay important work, then ways to overcome it. The final lecture links time to saying yes too easily. Agreeing to everything to please others quietly consumes the time needed for your own priorities.

> **Key takeaway:** Time is managed by choosing priorities, tackling the causes of delay, and not agreeing to everything.

WEEK 3 Assertiveness, Anger and Positive Thinking

Lectures: Types of people; How to say "no"; Controlling anger; Gaining power from positive thinking (two parts).

This week builds on the end of Week 2. It starts by describing different types of people you will meet at college and at work, so you can adjust how you deal with each. It then teaches how to say no politely but firmly, which is the practical side of assertiveness.

Emotions come next. The lecture on controlling anger covers how anger affects relationships and decisions and how to manage it instead of letting it manage you. The week closes with two lectures on positive thinking as a source of energy, confidence and resilience.

> **Key takeaway:** Being assertive, calm and positive protects your relationships and your wellbeing.

WEEK 4 Likeability and Personal Appeal

Lectures: What makes others dislike you; What makes others like you (two parts); Being attractive (two parts).

Week 4 looks at personality from the outside: how your habits and behaviour are received by others. It begins with the behaviours that make people dislike us, since avoiding them is often the quickest improvement, and then covers what makes others like us.

The last two lectures discuss being attractive in a wide sense. This goes beyond appearance to include manner, courtesy, confidence and the impression you leave in interactions. The aim is to understand these qualities so you can develop them deliberately.

> **Key takeaway:** Likeability comes from habits you can change: avoid what repels people and practise what draws them in.

WEEK 5 Common Errors in Communication

Lectures: Common Errors, parts 1 to 5.

Week 5 is a series of five lectures on common errors. The official syllabus gives only this title, with no sub-topics. The theme most likely covers frequent mistakes people make in language and expression, so this week is best read as a practical clinic on avoiding them.

Spreading the theme over five lectures suggests a gradual, example-driven approach, with each lecture handling a group of related mistakes. Fixing them improves clarity and credibility in emails, interviews, presentations and everyday conversation.

> **Key takeaway:** Small errors in language add up, so learning to spot and fix them quickly raises how professional you sound.

Based on the official NPTEL syllabus. This is a study summary, not a transcript of the lectures. Page 2

Summary of Lecture Notes

NPTEL Soft Skills course – weekly learning summaries (Weeks 1–8)

Enhancing Soft Skills and Personality | 8-Week Summary

WEEK 6 Humour and Money

Lectures: Humour in communication; Humour in the workplace; Function of humour in the workplace; Money and personality; Managing money.

The first three lectures treat humour as a serious communication skill. They cover how humour works in conversation, why it matters in the workplace, and what functions it serves, such as easing tension, building rapport and making ideas memorable. Used badly, humour can offend, so judgment about audience and context is part of the skill.

The second half connects money to personality. It looks at how attitudes toward money reflect character and then at managing money sensibly: earning, spending, saving and avoiding impulsive decisions.

> **Key takeaway:** Good humour builds relationships when used with care, and sound money habits are part of a mature personality.

WEEK 7 Health and Love

Lectures: Health and personality; Managing health 1: importance of exercise; Managing health 2: diet and sleep; Love and personality; Managing love.

Week 7 argues that personality cannot be separated from physical wellbeing. Three lectures cover health: why it matters for mood, focus and confidence, the importance of regular exercise, and the roles of diet and sleep. These everyday habits support everything else in the course.

The last two lectures turn to love and relationships. They link personality to the way people form close bonds and then discuss how to manage love and relationships in a healthy, balanced way.

> **Key takeaway:** Exercise, diet and sleep, along with healthy relationships, form the base for every other soft skill.

WEEK 8 Ethics, Etiquette, Memory and Environment

Lectures: Ethics and etiquette; Business etiquette; Managing mind and memory; Improving memory; Care for the environment; Highlights of the course.

The final week begins with ethics and etiquette: the values behind good behaviour and the manners that make interaction smooth. Business etiquette applies these to professional settings such as meetings, correspondence and workplace conduct.

It then turns to the mind, covering how to manage attention and improve memory, which is useful for studying and work. A lecture on caring for the environment widens the focus to responsibility beyond oneself. The course closes with a review of its main ideas.

> **Key takeaway:** A strong personality combines ethics, etiquette, a trained mind and responsibility toward others and the planet.

Putting the Course into Practice

- **Adopt a growth mindset.** Treat each new subject and each setback as a chance to improve through effort.
- **Plan your time.** Pick two or three priorities a day and start the hardest one first to limit procrastination.
- **Practise saying no.** Decline extra commitments politely so you can protect your main goals.
- **Audit your communication.** Note the language mistakes you repeat and correct them one at a time.
- **Protect your basics.** Keep a regular routine for exercise, food and sleep, and a simple monthly budget.

Sources: NPTEL course page and official syllabus for Enhancing Soft Skills and Personality (course 109104115, nptel.ac.in/courses/109104115). Week 5 lists no sub-topics in the syllabus, so its description is general.

SWOC Presentation

Career Roadmap: SWOT Analysis, 4-Year Plan and Presentation Script

CAREER ROADMAP | SWOT ANALYSIS | 4-YEAR PLAN

Career Roadmap: AI & Machine Learning Engineer

Specialisation: Machine Learning Engineering | Additional interest: Generative AI

First-year B.E. Computer Science and Engineering, St. Joseph's College of Engineering, Chennai

1. Executive Summary and Career Vision

The role of an AI-ML Engineer is to turn data into intelligent solutions. Unlike a standard software role, an AI-ML Engineer must combine programming, mathematics, data handling and problem-solving to build models that work on real-world problems. I am a first-year B.E. Computer Science and Engineering student at St. Joseph's College of Engineering, Chennai, and a beginner in this field, so this roadmap is written from a fresher's starting point.

My chosen domain is artificial intelligence and machine learning, with a specialisation in **machine learning engineering**: building, training, evaluating and deploying models. My additional area of interest is **generative AI**, including large language models. I come from a biology background in Class 12, so I will build my programming and mathematics foundations carefully, and I also see this background as a possible advantage later in areas such as healthcare AI. The plan below moves over four years from learning to code to building and deploying machine learning and generative AI applications, with a long-term path toward senior engineering roles.

2. SWOT Analysis (Honest Self-Assessment)

This analysis focuses on where I truly stand today. A beginner's strengths are mostly attitude, interest and potential, while the weaknesses are mostly missing skills that can be built step by step.

STRENGTHS	WEAKNESSES
• Strong interest in AI • Clear domain focus from year one • Started with C programming • Four full years ahead	• No projects or portfolio yet • Slow problem-solving speed • Biology-stream background • Python and ML tools still ahead
OPPORTUNITIES	**THREATS**
• High demand for AI, ML and GenAI • Free courses and free tools • Biology + AI niche • Strong generative AI ecosystem	• Rapid change in the field • Information overload • Strong competition • Over-reliance on AI tools

Strengths

- **Strong Interest in AI:** Genuine curiosity about AI and ML and a clear goal from the very first year, which will help me keep going through difficult topics.
- **Clear Direction:** I have already chosen my domain (AI and ML), my specialisation (machine learning engineer) and an additional interest (generative AI), so I can plan early instead of drifting.
- **A Start in Programming:** I have started with C programming, which builds the logic and discipline I need before learning Python and machine learning.
- **Time Advantage:** Four full years of study ahead to build skills, projects and experience without pressure.
- **Fresh Start:** Starting from the basics lets me learn the right fundamentals from good sources.

Weaknesses

- **No Projects Yet:** No completed projects, internships or GitHub portfolio so far, which means I cannot yet show practical proof of my skills.
- **Slow Problem-Solving Speed:** Coding problems take me a long time right now, and regular timed practice is needed to improve.

SWOC Presentation

Career Roadmap: SWOT Analysis, 4-Year Plan and Presentation Script

Career Roadmap: AI & Machine Learning Engineer

- **Biology-Stream Background:** Coming from biology in Class 12, I may have had less exposure to programming and mathematics-heavy subjects than classmates from other streams, so I must build these foundations deliberately.
- **Python and ML Tools Still Ahead:** So far I have started with C. Python, NumPy, Pandas, scikit-learn and Git are the next tools I need to learn.

Opportunities

- **Market Demand:** Growing demand for AI, machine learning and generative AI skills across almost every industry.
- **Free Learning Resources:** DeepLearning.AI's Machine Learning Specialization, NPTEL courses, Kaggle Learn, documentation and YouTube allow learning at low cost.
- **Free Tools:** Google Colab and Kaggle Notebooks give free computing, so I do not need expensive hardware to start.
- **Biology plus AI:** Healthcare, bioinformatics and drug discovery need people who understand both fields. My background could become a differentiator once my technical skills are strong.
- **Generative AI Ecosystem:** Open models, public APIs and frameworks let a beginner build real language-model applications.
- **Community and Events:** Hackathons, coding contests, college technical clubs and open-source projects welcome beginners.

Threats

- **Rapid Change:** AI changes quickly, so learning must continue throughout my career.
- **Information Overload:** So many tools and courses exist that a beginner can feel lost without a clear plan.
- **Competition:** Many students are targeting AI-ML roles, so projects and practical proof are needed to stand out.
- **Consistency:** Balancing college studies, exams and self-learning requires discipline over four years.
- **Over-Reliance on AI Tools:** Generative AI can write code for me, which could weaken my own fundamentals if I use it to skip the learning instead of supporting it.

3. Tech Action Plans

Each weakness found in the SWOT analysis is matched below with a concrete action and a measurable target.

Weakness	Action	Target
No projects yet	Create a GitHub account in the first month and upload all practice code. Build small Python projects first, then ML projects.	1 to 2 small projects in Year 1; 2 to 3 ML projects by Year 2
Slow problem solving	Solve 1 to 2 problems a day, starting with easy ones. Time myself, then review better solutions and note the patterns.	100+ problems in the first year
Biology background and maths	Rebuild school-level maths first, then study linear algebra, probability, statistics and basic calculus for ML using free courses.	Core maths for ML covered in 6 to 12 months
Python and ML tools	Learn Python after C basics, then NumPy, Pandas, Matplotlib, scikit-learn and Git.	Beginner Python course finished in Year 1

Focus areas

- **Programming and Problem-Solving (immediate priority):** Strengthen C, learn Python, and study data structures and algorithms.
- **Mathematics for ML:** Study linear algebra, probability, statistics and basic calculus so that I understand why algorithms work, not only how to call them.
- **Machine Learning Fundamentals:** Follow a structured course such as the Machine Learning Specialization, practising with NumPy, Pandas and scikit-learn.
- **Portfolio and Visibility:** Upload all work to GitHub, join Kaggle competitions and take part in hackathons, so that I always have visible proof of my skills.

SWOC Presentation

Career Roadmap: SWOT Analysis, 4-Year Plan and Presentation Script

Career Roadmap: AI & Machine Learning Engineer

- **Generative AI Track (from Year 2):** Learn how large language models work, then prompt engineering, calling model APIs, retrieval-augmented generation (RAG) and the basics of fine-tuning, and build one or two small applications.
- **ML Engineering and Deployment:** Learn Git, Docker, a web framework such as FastAPI or Flask and a cloud platform, so that my models run as real services, not only in notebooks.
- **Communication and Teamwork:** Write short blog posts, explain what I learn to classmates and work on group projects.
- **AI Ethics and Responsible AI:** Study data privacy, bias and fairness by Year 3 so that the systems I build are safe and trustworthy.

4. 4-Year Academic and Professional Milestones

A structured progression from beginner learner to a job-ready ML engineer with generative AI skills.

Year 1: Foundation Building (The "Learner" Phase)	
Academic focus	Programming (C and Python), data structures and algorithms basics, and mathematics foundations (algebra, basic calculus, probability, statistics).
Technical milestones	Solve 100+ coding problems; create a GitHub account and upload all practice code; complete 1 to 2 small Python projects; finish a beginner AI/ML introduction course.
Certification	A beginner Python course certificate or Google AI Essentials.

Year 2: Core Machine Learning (The "Practitioner" Phase)	
Academic focus	Machine learning (supervised and unsupervised learning, feature engineering, model evaluation), linear algebra and statistics in depth, and Python libraries (NumPy, Pandas, Matplotlib, scikit-learn).
Technical milestones	Complete 2 to 3 ML projects with real datasets; take part in 1 to 2 Kaggle competitions or hackathons; apply for a summer internship or mini-project in AI/ML; try first small experiments with a language-model API.
Certification	Machine Learning Specialization (DeepLearning.AI) or equivalent.

Year 3: Deep Learning, Generative AI and Deployment (The "Builder" Phase)	
Academic focus	Deep learning (neural networks, CNNs, RNNs, Transformers), NLP and computer vision basics, PyTorch or TensorFlow, Git and Docker, plus large language models, prompt engineering and RAG.
Technical milestones	Build and deploy a model as a web app or API using FastAPI or Flask; build a generative AI application such as a document question-answering chatbot; complete an AI/ML internship; publish a technical blog post; make a first open-source contribution.
Certification	Deep Learning Specialization and Generative AI with Large Language Models (DeepLearning.AI and AWS); optionally AWS Certified Cloud Practitioner.

Year 4: Professionalisation and Placement (The "Engineer" Phase)	
Academic focus	A capstone project that solves a real-world problem with ML or generative AI (for example, one linked to healthcare or biology), and electives such as AI Ethics, Big Data or Generative AI and LLMs; basics of model monitoring and versioning.
Technical milestones	Maintain a GitHub portfolio with 4 to 5 quality projects; build an active LinkedIn profile; attend AI conferences or workshops; practise technical and HR mock interviews.
Outcome	Secure a role as a Junior ML Engineer, AI Engineer or Data Scientist, or an intern-to-full-time position, with a clear path toward Senior ML Engineer and Generative AI Engineer roles.

5. Conclusions and Key Takeaways

- **Fundamentals First:** Strong programming, mathematics and problem-solving come before advanced AI topics.
- **Consistency Beats Speed:** Small, regular practice over four years will take me further than short bursts of effort.

SWOC Presentation

Career Roadmap: SWOT Analysis, 4-Year Plan and Presentation Script

Career Roadmap: AI & Machine Learning Engineer

- **Projects Over Certificates:** A GitHub portfolio of working projects proves skill better than certificates alone.
- **Use AI Tools to Learn, Not to Skip Learning:** Generative AI is a powerful helper, but I must still understand the code and the maths myself.
- **My Background Is a Starting Point:** Coming from biology means more work on foundations at first, and possibly a useful edge in healthcare and life-science AI later.
- **Adaptability Is Survival:** AI changes fast, so continuous learning must become a lifelong habit.

6. Presentation Script

How to use this script: Make one slide for each part below. Put the slide title and the short points under "On the slide" on the slide itself, then say the "What to say" part aloud in your own words. It takes about 5 to 7 minutes in total.

Slide 1: My Journey to Becoming a Machine Learning Engineer

On the slide: Career roadmap with a Generative AI interest • First-year B.E. CSE, St. Joseph's College of Engineering, Chennai

What to say: *"Good day, everyone. I am a first-year B.E. Computer Science and Engineering student at St. Joseph's College of Engineering, Chennai. Today I will share my career roadmap. My goal is to become a machine learning engineer, and I am also interested in generative AI."*

Slide 2: Why Machine Learning Engineering and Generative AI

On the slide: ML engineers build, train, test and deploy models • Generative AI: models that write text and answer questions • High demand across industries

What to say: *"A machine learning engineer does not only build a model. They also train it, test it and make it work in a real product. That is why I chose this path. I am also interested in generative AI, such as language models that can write and answer questions, and I want to learn how they work, not only how to use them."*

Slide 3: Where I Stand Today: My SWOT Analysis

On the slide: Strengths: interest in AI, clear goal, C programming • Weaknesses: no projects, slow problem solving, biology background • Opportunities: free courses, high demand, biology + AI • Threats: fast change, competition, over-reliance on AI tools

What to say: *"Before planning, I checked honestly where I stand. My strengths are my interest in AI, a clear goal and my start in C programming. My weaknesses are that I have no projects yet, I solve problems slowly, and I come from a biology background, so I must build my programming and maths carefully. The opportunities are free courses and tools and the growing demand for AI. The threats are fast change, strong competition and depending too much on AI tools."*

Slide 4: My Action Plan

On the slide: Solve 100+ coding problems in Year 1 • Learn Python, maths for ML and core ML libraries • Put every project on GitHub • Add a generative AI track from Year 2

What to say: *"To fix my weaknesses, I made an action plan. In the first year, I will solve at least 100 coding problems, learn Python, and upload all my code to GitHub. I will also rebuild my maths for machine learning. Then I will learn machine learning with NumPy, Pandas and scikit-learn and build two to three projects by the end of the second year. From Year 2, I will add generative AI and learn to deploy my models."*

Slide 5: My 4-Year Roadmap

On the slide: Year 1: Learner (coding and maths) • Year 2: Practitioner (core machine learning) • Year 3: Builder (deep learning, GenAI, deployment) • Year 4: Engineer (capstone and placement)

What to say: *"My roadmap has four phases. In Year 1, I build my foundations in coding and maths. In Year 2, I learn core machine learning and complete my first projects. In Year 3, I move to deep learning, generative AI and deployment, and I aim for an internship. In Year 4, I complete a capstone project, build my GitHub portfolio and prepare for placement."*

Slide 6: Conclusion and Thank You

On the slide: Fundamentals first, then advanced AI • Consistent practice and real projects • Thank you. Any questions?

What to say: *"To conclude, becoming a machine learning engineer is a marathon, not a sprint. It needs strong basics, regular practice and real projects. I am a beginner today, but with a clear plan and steady effort, I believe I can reach my goal. Thank you for listening. I am happy to answer your questions."*

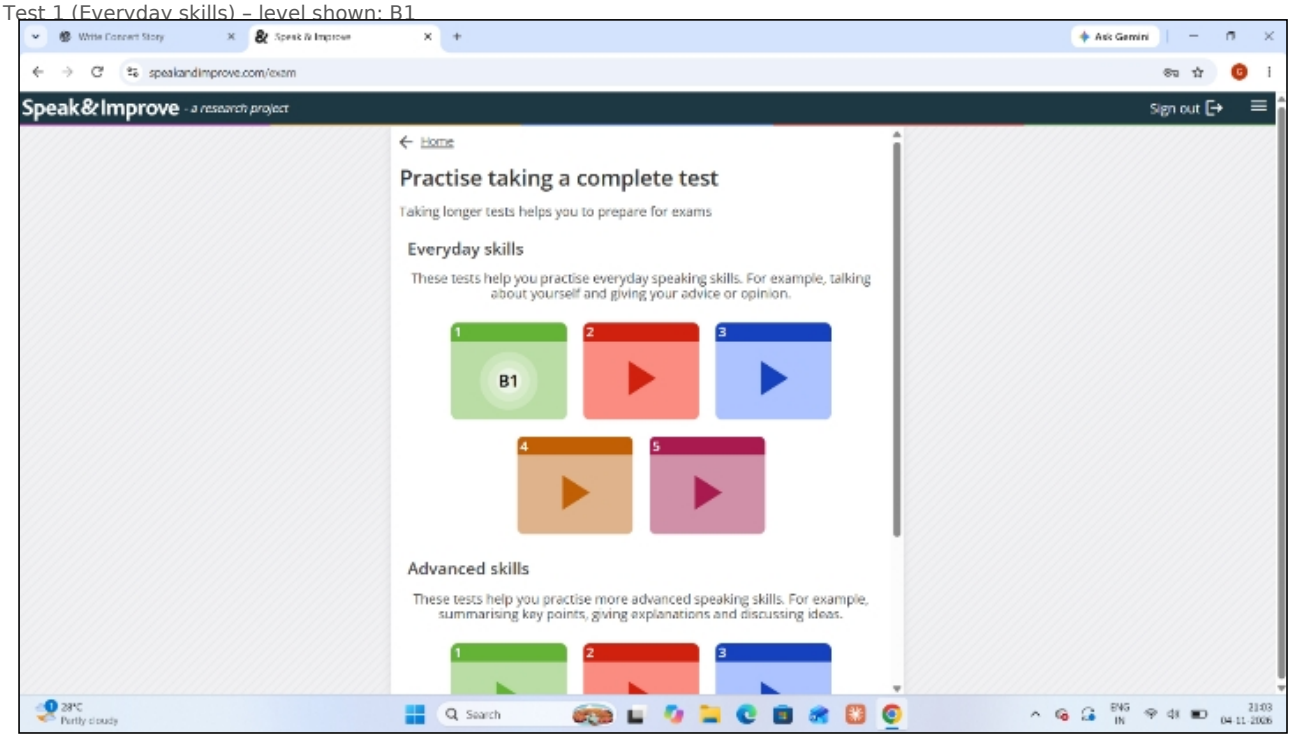
Note: The courses named above were confirmed to exist in October 2026. Check the official pages before enrolling.

Write and Improve / Speak and Improve

Cambridge Speak & Improve and Write & Improve – result screenshots

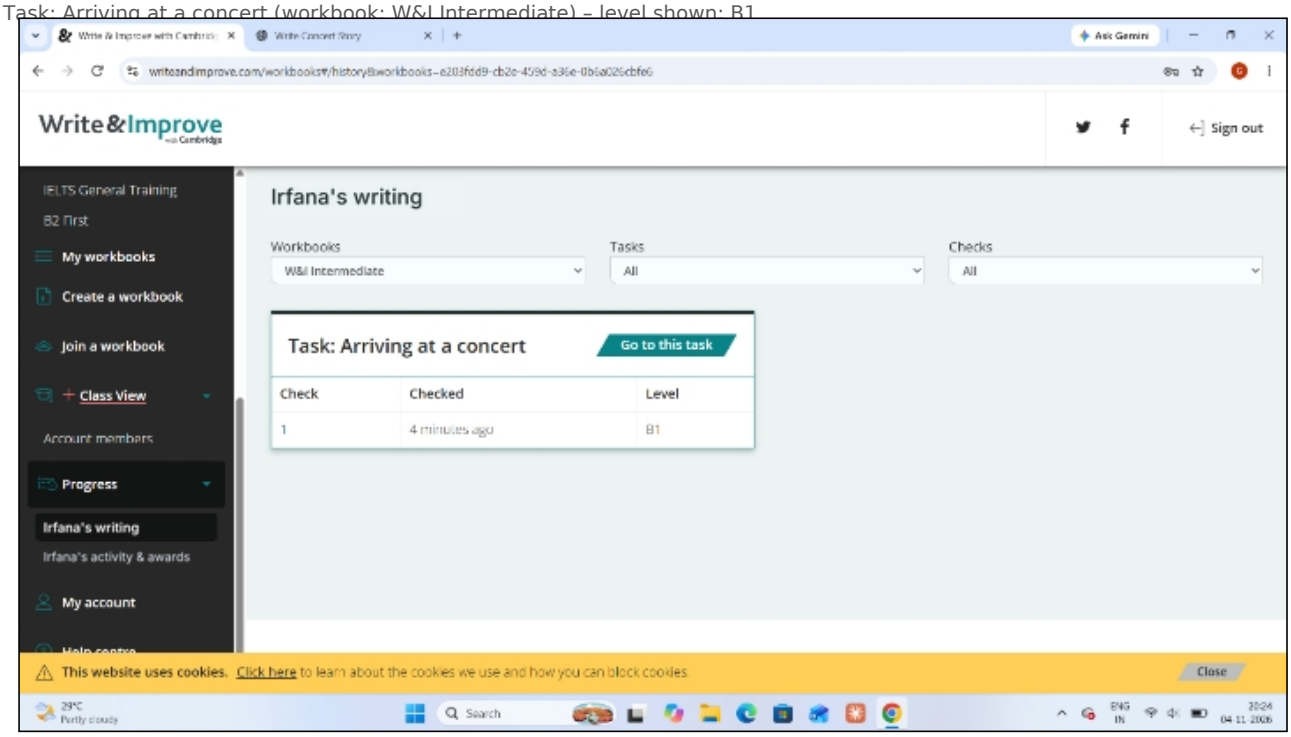
Speak & Improve: complete test result

Test 1 (Everyday skills) – level shown: B1



Write & Improve: Irfana's writing

Task: Arriving at a concert (workbook: W&I Intermediate) – level shown: B1



Professional English Practice Worksheets – Solved

Course Code: EN25C09 | Subject: Professional English – I (Lab Integrated)
Worksheet 1: Model Exam I (Parts A, B, C), followed by Worksheet 2: Model – I Practice Worksheet.

St. Joseph's College of Engineering
OMR, Chennai - 119 (An Autonomous Institution)
Professional English - I (Lab Integrated) | EN25C09
Model I Practice Worksheet - Questions and Model Answers

PART A - Short Grammar Questions

15 questions | 1 mark each

1. Fill in the blank with the correct form of the verb: The automated system ___ (monitor) the temperature continuously to ensure safe operation.

Ans: monitors.

2. Fill in the blank with the correct past form: The engineers ___ (record) the vibration readings during yesterday's trial.

Ans: recorded.

3. Use the comparative form: The newly developed composite material is significantly ___ (strong) than the conventional material while maintaining a lower weight.

Ans: stronger.

4. Choose must/can/might: Engineers ___ follow safety regulations while operating heavy machinery.

Ans: must.

5. Choose a modal expressing capability: Modern sensors ___ detect even minor changes in temperature.

Ans: can.

6. Combine using a suitable relative pronoun: The laboratory developed a new testing device. The device measures material strength with high precision.

Ans: The laboratory developed a new testing device that measures material strength with high precision.

7. First conditional: If the temperature ___ (rise) above the permissible limit, the cooling system will activate automatically.

Ans: rises.

8. Third conditional: If the technician had inspected the circuit earlier, the equipment ___ (not / fail) during operation.

Ans: would not have failed.

9. Combine using an infinitive of purpose: The mechanics installed dual damper springs. They wanted to isolate chassis resonance.

Ans: The mechanics installed dual damper springs to isolate chassis resonance.

10. Use the gerund after the preposition: The research team reduced energy consumption by ___ (optimize) the motor control algorithm.

Ans: optimizing.

11. Convert to impersonal passive: The lab assistant poured 50 mL of hydrochloric acid into the titration flask.

Ans: Fifty millilitres of hydrochloric acid was poured into the titration flask.

12. Use the adverb from smooth: The robotic arm moves ___ to ensure accurate positioning.

Ans: smoothly.

13. Rewrite in indirect speech: The project supervisor asked the engineer, "Did you calibrate the pressure sensor before testing?"

Ans: The project supervisor asked the engineer whether he or she had calibrated the pressure sensor before testing.

14. Correct the noun form: The laboratory purchased several new equipments for conducting material testing.

Ans: The laboratory purchased several new pieces of equipment for conducting material testing.

15. Use the correct irregular comparative: Because the heat sink had poor thermal paste contact, heat dissipation was ___ (bad) than expected.

Ans: worse.

PART B - Applied Grammar and Writing

10 questions | 2 marks each

1. Rewrite in the simple past: The engineer calibrates the pressure sensor and verifies its accuracy.

Ans: The engineer calibrated the pressure sensor and verified its accuracy.

2. Draft two comparative sentences comparing a laptop and desktop in portability and screen size.

Ans: Portability: A laptop is more portable than a desktop computer.
Screen size: A desktop computer usually has a larger screen than a laptop.

3. Combine with a relative clause: The engineer designed a new cooling system. It reduces the temperature of industrial machines.

Ans: The engineer designed a new cooling system that reduces the temperature of industrial machines.

4. Complete logically: (a) If the battery voltage drops below the required level, ... (b) If the technician had checked the connections earlier, ...

Ans: (a) If the battery voltage drops below the required level, the warning indicator will light up.
(b) If the technician had checked the connections earlier, the equipment might not have stopped.

5. Convert to impersonal passive: The laboratory assistant heated the solution to 80 degrees C before testing.

Ans: The solution was heated to 80 degrees C before testing.

6. Rewrite in indirect speech: The project manager asked the technician, "Have you inspected the pressure valves before starting the compressor?"

Ans: The project manager asked the technician whether he or she had inspected the pressure valves before starting the compressor.

7. Correct the mass noun errors: (a) The engineer received many feedbacks on the new design. (b) The students received several useful advices from their teacher.

Ans: (a) The engineer received a lot of feedback on the new design. (b) The students received several useful pieces of advice from their teacher.

8. Rewrite using an infinitive of purpose or gerund: (a) The technician wears safety gloves. He wants to protect his hands. (b) The cooling fan operates continuously. It prevents the motor from overheating.

Ans: (a) The technician wears safety gloves to protect his hands. (b) The cooling fan operates continuously to prevent the motor from overheating.

9. Use the correct adverb: (a) The machine operates ___ (efficient). (b) The technician checked the electrical connections ___ (careful).

Ans: (a) efficiently (b) carefully.

10. Insert modal auxiliaries for capability and strict prohibition: (a) The robotic arm ___ lift heavy components. (b) Visitors ___ enter the testing area without authorization.

Ans: (a) can (b) must not.

PART C - Extended Writing

The worksheet asks you to answer Question 26 and choose one alternative in Questions 27-30. Both alternatives are provided here for practice.

Question 26 - Internet Safety Instructions

Prompt: Write eight imperative safety instructions for safe and responsible internet use.

Model Answer

1. Use a strong, unique password for every important account.
2. Turn on two-factor authentication whenever it is available.
3. Do not click suspicious links or open unexpected attachments.
4. Verify a website address before entering personal or payment information.
5. Keep your operating system, browser and security software updated.
6. Never share passwords, one-time codes or sensitive personal details with strangers.
7. Use secure networks and avoid sensitive transactions on public Wi-Fi.
8. Report suspicious messages, cyberbullying and suspected account activity to a trusted adult or the relevant service.

Question 27.a - Traditional Art in Tamil Nadu

Prompt: Write about a regional traditional art form, its origins and practices, and ways to preserve it.

Model Answer

Therukoothu, or street theatre, is a traditional performance art of Tamil Nadu. It grew in village communities as a way to share stories, celebrate festivals and bring people together. Many performances draw on episodes from the Mahabharata and other well-known stories. The open-air setting made the art accessible to everyone, without the need for a formal theatre.

A traditional performance combines acting, music, dance, colourful costumes and expressive makeup. Performers sing and speak in Tamil, while drums and other instruments support the rhythm. The actors often use energetic gestures and direct interaction with the audience. Performances may continue for several hours, especially during temple festivals. Artists learn through practice, observation and guidance from experienced performers.

Today, migration, changing entertainment habits and limited income can make it difficult for younger people to continue this demanding art. Schools and colleges can invite performers for demonstrations and workshops. Local cultural groups can record performances, interview senior artists and create subtitled videos for online audiences. Festival organisers can provide fair fees, safe stages and regular performance opportunities. Grants and scholarships can support young learners.

Modern technology can help preserve Therukoothu when it documents and promotes the art without replacing live performance. Young people can learn its music, stories and stage skills while also sharing it through responsible digital media. Supporting the artists and giving them opportunities to teach will keep this cultural tradition meaningful for future generations.

Question 27.b - Educational Trip to Thanjavur

Prompt: Write a structured 250-word travel plan and feature about a historical and cultural place.

Model Answer

Our class will take an educational trip to Thanjavur, a historic city in Tamil Nadu known for its temples, art and Chola heritage. We will travel by train, meet at the college early in the morning and use a hired bus between the railway station and the main sites. Students will carry water, identification and notebooks, and follow the teacher's group instructions.

Our first visit will be the Brihadisvara Temple, a UNESCO World Heritage Site built during the Chola period. We will observe its grand tower, stone sculptures and inscriptions, while our guide explains the engineering and religious history of the complex. We will then visit the Thanjavur Maratha Palace and its art gallery to learn about royal life, manuscripts and regional painting. The Saraswathi Mahal Library area will help us understand the importance of preserving historical records.

At lunch, we will try local food such as sambar, rice, vegetable curries and traditional sweets, choosing a clean restaurant and avoiding food waste. We will speak politely with local people and notice the region's Tamil language, temple customs, craft traditions and hospitality. Students will sketch one architectural detail or write a short observation during the visit.

We will return by evening after checking that everyone is present and leaving each site clean. The trip will connect classroom learning with real places and objects. It will also show us how history, architecture, food and daily traditions shape a region's identity. We will prepare a group report and share what we learned with our classmates.

Question 28.a - Request to Extend Library Hours

Prompt: Write a formal letter to the college librarian requesting longer hours during semester examinations.

Model Answer

From:
A. Student
Second Year, Department of Engineering
[College Name]
4 November 2026

To:
The Librarian
[College Name]

Subject: Request to extend library working hours during semester examinations

Respected Sir/Madam,

I am writing on behalf of students in our department to request an extension of the college library's working hours during the forthcoming semester examination period. Many students need a quiet place to revise, consult reference books and work on assignments after regular classes. Longer access would particularly help students who cannot study effectively in crowded or noisy accommodation.

We expect many students to use the library, especially in the evenings. We propose extending the closing time by two hours on weekdays during the examination period, subject to staff availability and college approval. Students can sign in with their college identification cards, maintain silence, return books on time and follow all library rules.

The library may keep the reference and study areas open, with a staff member or security officer present. A sign-up sheet can help estimate attendance, and students can volunteer to leave their study spaces tidy. We understand that staffing and safety must be considered, and we would be grateful for any schedule the college can support.

This arrangement would encourage focused study and equal access to academic resources. Thank you for considering our request.

Yours faithfully,
A. Student
Class Representative

Question 28.b - Request to Attend an AI Workshop

Prompt: Write a formal letter to the principal requesting permission to attend a technical workshop.

Model Answer

From:
A. Student
Second Year, Department of Engineering
[College Name]
4 November 2026

To:
The Principal
[College Name]

Subject: Request for permission to attend a workshop on Artificial Intelligence

Respected Sir/Madam,

I respectfully request permission to participate in a technical workshop on Artificial Intelligence and Emerging Technologies organised by [Institution Name] on [date] at [venue]. The programme is intended for engineering students and includes expert talks, demonstrations and practical sessions.

I wish to understand current AI applications, learn the basics of responsible model development and explore how emerging technologies can support engineering work. I expect to improve my technical awareness, ask questions of industry and academic experts, and gain ideas for future projects. After the workshop, I will prepare a short report and share the key learning points with my classmates.

I will register only after receiving the college’s approval. I will submit the programme details and fee information for review, arrange safe transport, attend all sessions and follow the organiser’s rules. I will also ensure that my classwork is completed and obtain any notes or assignments missed during my absence. If the college requires a faculty member to accompany participants, I will follow that arrangement.

I request your permission and, if possible, an official letter confirming my student status. Thank you for considering my application.

Yours obediently,
A. Student
Second Year, Department of Engineering

Question 29.a - AI-Based and Teacher-Based Assessment

Prompt: Write a 250-300 word analytical comparison addressing accuracy, feedback time, fairness and human judgement.

Model Answer

Assessment helps engineering students understand what they know and what they need to practise. Artificial Intelligence-based assessment and traditional teacher-based assessment can both support learning, but each has strengths and limitations.

AI systems can mark objective questions quickly and apply the same scoring rules to every submission. They can analyse large amounts of work, identify patterns in errors and provide feedback soon after a student completes a task. This speed is useful in large classes, where a teacher may need several days to return work. Automated tools can also offer extra practice tailored to a student's weaknesses.

However, accuracy depends on the quality of the questions, answer keys and training data. An AI system may misread an unusual but valid explanation, reward formulaic responses or perform poorly when students use unfamiliar language. If its design reflects biased data, its results may be unfair to some learners. Students also need clear information about how their work is collected and assessed.

Teachers can interpret context, recognise original reasoning and ask follow-up questions. They can judge laboratory teamwork, design choices and communication in ways that are difficult to reduce to a score. Their feedback may take longer, and personal expectations can sometimes create inconsistency, but professional judgement allows them to explain a decision and respond to a student's needs.

A balanced approach is most effective. AI can assist with routine marking, provide practice and flag work for review. Teachers should set assessment goals, check automated results and make final decisions on complex work. In engineering education, accuracy and fairness require both consistent tools and human oversight. Used transparently, AI can save time while teachers preserve the judgement, encouragement and accountability that good assessment needs.

Question 29.b - Urban and Rural Living

Prompt: Write a 250-300 word compare-and-contrast essay about employment, cost, services and quality of life.

Model Answer

Choosing between urban and rural life depends on a person's work, family needs and preferred surroundings. Both settings offer valuable opportunities, but they differ in employment, cost, services and pace of life.

Cities usually provide a wider range of jobs in engineering, healthcare, business and technology. They also have more colleges, training centres and specialist hospitals. Public transport and digital services may make it easier to reach work and education. Yet urban housing, transport and daily expenses can be high. Heavy traffic, crowded spaces and noise may reduce leisure time and contribute to stress.

Rural areas often have lower housing costs, closer contact with nature and stronger ties among neighbours. Farming, local businesses, crafts and public services provide employment, although the number of specialised jobs may be limited. Schools and clinics are available in many rural communities, but advanced courses or specialist healthcare can require long travel. Reliable transport and internet access also vary between places.

Quality of life is not determined by location alone. Some people value the variety and convenience of a city, while others prefer a quieter environment and more space. Rural communities can benefit from improved transport, broadband and investment in schools and healthcare. Cities can improve living conditions by expanding affordable housing, green spaces and efficient public transport.

In conclusion, urban areas offer greater access to jobs and specialised services, while rural areas often provide a lower-cost and calmer way of life. The best choice depends on individual priorities. With thoughtful planning and fair access to essential services, both settings can support healthy, fulfilling lives.

Question 30.a - Email About First Semester

Prompt: Write a personal email to a school friend about your first semester in engineering.

Model Answer

To: friend@example.com
Subject: My first semester in engineering

Hi Ravi,

How are you? I have been meaning to tell you about my first semester at engineering college. It has been busy, exciting and very different from school.

I have made friends with classmates from several towns. We help one another with notes and work together on small lab tasks. The classroom activities include lectures, demonstrations, group discussions and practical sessions. I especially enjoy seeing how ideas from physics and mathematics are used in experiments.

Adjusting to college life took a few weeks. I had to plan my day, keep track of deadlines and become more independent. The campus is much larger than our school, and I am still discovering useful places to study and relax.

Our semester examinations are coming soon, so I have started revising each subject and solving sample problems. I feel a little nervous, but my study group keeps me motivated. I hope the exams go well and that the next semester brings new projects and experiences.

Write back and tell me how you are doing. I hope we can meet during the holidays.

Best wishes,
Arun

Question 30.b - Email About Orientation

Prompt: Write a personal email describing your first-year engineering orientation programme.

Model Answer

To: friend@example.com
Subject: College orientation was a great start

Hi Ravi,

I hope you are well. I attended the orientation programme for first-year engineering students last week, and it helped me feel more prepared for college.

The day began with a welcome from the principal and introductions to the department heads. Senior students led a campus tour and showed us the classrooms, laboratories, library, computer centre and student support office. We also attended short sessions on study skills, clubs and the academic calendar.

The staff explained important college regulations, including attendance, laboratory safety, library conduct and respectful behaviour. The laboratory demonstration was especially useful because it showed us how to handle equipment carefully and where to find safety instructions. There were also group activities that helped me talk to other new students. I have already made a few friends.

Before the programme, I was unsure about how college classes would work and whom to ask for help. Now I know where to find resources and how to contact my teachers. The orientation made the campus feel more welcoming and gave me a clearer idea of the responsibilities of an engineering student.

I am looking forward to classes and practical work. Tell me how your studies are going, and let us catch up soon.

Your friend,
Arun

St. Joseph's College of Engineering

OMR, Chennai – 119 (An Autonomous Institution)

Professional English – I (Lab Integrated)

Model – I – Practice Worksheet: Questions with Answers

Course Code: EN25CO9

PART A

Exercise 1: Simple Present Tense (Subject-Verb Concord)

Fill in the blanks with the correct form of the verbs given in brackets.

1. Neither the project supervisor nor the lead programmers _____ (recognize / recognizes) the anomaly in the build script.
   Ans: recognize – with neither…nor, the verb agrees with the nearer subject ("programmers", plural).
2. Each of the automated sorting arms _____ (require / requires) recalibration after twelve hours of continuous duty.
   Ans: requires – "Each" is singular.
3. Either the primary sensor or the auxiliary telemetry units _____ (transmit / transmits) packet data to the ground station.
   Ans: transmit – the nearer subject "units" is plural.
4. The consortium of aerospace contractors _____ (agree / agrees) to adhere to the revised tolerance levels.
   Ans: agrees – "consortium" is a collective noun acting as one unit.
5. Physics and material science _____ (form / forms) the core theoretical foundation for nanoscale fabrication.
   Ans: form – two subjects joined by "and" take a plural verb.

Exercise 2: Simple Past Tense (Trial Reporting)

Fill in the blanks with the simple past form of the given verbs.

1. Last month, the metallurgy division _____ (conduct) destructive fatigue testing on the titanium alloy shaft.
   Ans: conducted
2. The field engineers _____ (deploy) thirty IoT water-monitoring nodes across the river basin.
   Ans: deployed
3. During the pressure overload trial, the relief valve _____ (open) at 150 psi as designed.
   Ans: opened
4. The research unit _____ (publish) their initial benchmark results on low-power RISC-V architectures.
   Ans: published
5. In the 2025 dry run, the autonomous navigation algorithm _____ (detect) all dynamic road obstacles successfully.
   Ans: detected

Exercise 3: Comparative Adjectives (Technical Metrics)

Complete each sentence with the appropriate comparative form of the adjective in brackets.

1. Gallium nitride (GaN) semiconductors operate at _____ (high) power efficiency than standard silicon transistors.
   Ans: higher
2. Solid-state batteries deliver a _____ (safe) operational threshold under rapid discharge cycles than traditional lithium cells.
   Ans: safer
3. The revised compiler generates code that executes _____ (fast) than the legacy unoptimized routine.
   Ans: faster
4. Brushless DC motors are considerably _____ (durable) than brushed equivalents due to reduced mechanical wear.
   Ans: more durable
5. The liquid cooling loop maintained a _____ (low) core temperature than the dual fan-sink array.
   Ans: lower

Exercise 4: Modals of Obligation & Necessity

Fill in each blank with an appropriate modal auxiliary (must, should, or have to / has to).

1. Every operator _____ wear anti-static wrist straps before handling open CMOS motherboards.
   Ans: must (also: has to)
2. Hazardous chemical waste _____ be neutralized prior to disposal into the industrial drain network.
   Ans: must (also: has to)
3. Field technicians _____ log their entry time into the high-voltage testing bay without fail.
   Ans: have to (also: must)
4. To maintain ISO certification, calibration logs _____ be audited biannually by an external inspector.
   Ans: must (also: have to)
5. The test rig _____ undergo an emergency stop inspection before initiating the high-speed turbine run.
   Ans: must (also: has to)

Exercise 5: Modals of Capability & Possibility

Fill in each blank with a suitable modal (can, could, may, might).

1. With the upgraded DSP chip, the system _____ process 4K video streams in real time.
   Ans: can
2. If thermal throttling occurs, the processing clock rate _____ drop by as much as 30%.
   Ans: may / might / could – possibility.
3. Advanced laser profilometry _____ measure surface roughness down to sub-micron tolerances.
   Ans: can
4. Unshielded signal lines _____ pick up electromagnetic interference from nearby induction coils.
   Ans: can / may
5. The modular chassis design _____ accommodate up to four secondary expansion cards.
   Ans: can

Exercise 6: Relative Clauses

Combine each pair of sentences into a single sentence using a suitable relative pronoun (who, which, that, whose).

1. (a) A civil engineer designed the suspension cable bridge. (b) She received the national infrastructure award.
   Ans: The civil engineer **who** designed the suspension cable bridge received the national infrastructure award.
2. (a) The spectrometer analyzes infrared emission spectra. (b) It is mounted inside the cleanroom vacuum chamber.
   Ans: The spectrometer, **which** analyzes infrared emission spectra, is mounted inside the cleanroom vacuum chamber.
3. (a) The technical specialist was reassigned to the robotics lab. (b) His patent on pneumatic grippers was approved.
   Ans: The technical specialist **whose** patent on pneumatic grippers was approved was reassigned to the robotics lab.
4. (a) This microcontroller features ultra-low power sleep states. (b) It is ideal for battery-operated remote sensors.
   Ans: This microcontroller, **which** features ultra-low power sleep states, is ideal for battery-operated remote sensors.
5. (a) The field researchers surveyed the wind turbine farm. (b) They submitted their acoustic emission assessment yesterday.
   Ans: The field researchers **who** surveyed the wind turbine farm submitted their acoustic emission assessment yesterday.

Exercise 7: If-Clauses (Type 1 – Real Conditionals)

Complete the sentences with the correct form of the verbs in brackets.

1. If the bus voltage _____ (exceed) 24V, the zener diode will divert the surge to ground.
   Ans: exceeds
2. The watchdog timer will reset the CPU if the main thread _____ (fail) to ping it within 500 ms.
   Ans: fails
3. If the operator _____ (engage) the safety latch, the hydraulic press will not actuate.
   Ans: engages
4. If temperature levels _____ (rise) above 90°C, the cooling fans will switch to maximum RPM.
   Ans: rise
5. The system will trigger an auditory alarm if the toxic gas detector _____ (register) more than 20 ppm.
   Ans: registers

Exercise 8: If-Clauses (Type 3 – Unreal Past / Troubleshooting)

Fill in the blanks with the past unreal conditional form (had + past participle / would have + past participle).

1. If the backup generator _____ (start) immediately, the data centre servers would not have shut down.
   Ans: had started
2. The turbine bearing would not have seized if the maintenance crew _____ (lubricate) it as scheduled.
   Ans: had lubricated
3. If the security team had patched the zero-day exploit earlier, the data breach _____ (prevent).
   Ans: would have been prevented
4. The circuit breaker _____ (trip) safely if the overcurrent sensor had functioned properly.
   Ans: would have tripped

5. If the telemetry link _____ (remain) stable, ground engineers would have corrected the satellite's orbital trajectory.
   Ans: had remained

Exercise 9: Infinitives of Purpose

Combine the two ideas using an infinitive of purpose (to + verb).

1. (a) The mechanics installed dual damper springs. (b) They wanted to isolate chassis resonance.
   Ans: The mechanics installed dual damper springs **to isolate** chassis resonance.
2. (a) Technicians purge the fuel line with dry nitrogen gas. (b) They want to remove residual flammable vapors.
   Ans: Technicians purge the fuel line with dry nitrogen gas **to remove** residual flammable vapors.
3. (a) We introduced an optical encoder. (b) We intended to determine spindle velocity accurately.
   Ans: We introduced an optical encoder **to determine** spindle velocity accurately.
4. (a) The developers wrote an automated load testing script. (b) Their goal was to simulate ten thousand concurrent connections.
   Ans: The developers wrote an automated load testing script **to simulate** ten thousand concurrent connections.
5. (a) Gold plating is applied to terminal connectors. (b) Its purpose is to prevent atmospheric corrosion.
   Ans: Gold plating is applied to terminal connectors **to prevent** atmospheric corrosion.

Exercise 10: Gerunds after Prepositions

Fill in the blanks with the gerund form of the bracketed verb after the preposition.

1. The aerodynamic team improved aircraft fuel economy by _____ (redesign) the winglet profile.
   Ans: redesigning
2. Always disconnect the power supply before _____ (replace) defective surface-mount components.
   Ans: replacing
3. The facility was praised for _____ (reduce) volatile organic chemical emissions by 40%.
   Ans: reducing
4. Avoid _____ (clean) precision optical lenses with abrasive industrial solvents.
   Ans: cleaning (“avoid” takes a gerund directly).
5. The digital signal processor isolates target audio signals by _____ (filter) out background ambient noise.
   Ans: filtering

Exercise 11: Impersonal Passive Voice

Convert the following active laboratory/industrial actions into the Impersonal Passive Voice.

1. The lab assistant poured 50 mL of hydrochloric acid into the titration flask.
   Ans: 50 mL of hydrochloric acid **was poured** into the titration flask.
2. The automation unit tested the robotic arm under a 50 kg dynamic payload.
   Ans: The robotic arm **was tested** under a 50 kg dynamic payload.
3. Field engineers calibrated the optical pyrometer prior to ignition.
   Ans: The optical pyrometer **was calibrated** prior to ignition.
4. The technicians observed severe cavitation damage along the impeller blades.
   Ans: Severe cavitation damage **was observed** along the impeller blades.
5. We store volatile solvents in flame-resistant containment cabinets.
   Ans: Volatile solvents **are stored** in flame-resistant containment cabinets.

Exercise 12: Adverbs of Manner

Fill in the blanks with the appropriate adverb derived from the adjective in brackets.

1. The stepper motor rotates _____ (precise) in 1.8-degree angular increments.
   Ans: precisely
2. The telemetry ground station _____ (periodic) polls all deployed marine buoy sensors.
   Ans: periodically
3. The thermal imaging camera registered the temperature drop _____ (rapid) during quenching.
   Ans: rapidly
4. Structural joints must be welded _____ (firm) to withstand heavy cyclic vibrational stress.
   Ans: firmly
5. The autonomous drone traversed the confined pipeline corridor _____ (smooth).
   Ans: smoothly

Exercise 13: Reported Speech (Interrogative Structures)

Transform each direct technical query into indirect/reported speech.

1. The quality manager asked the technician, “Has the insulation resistance test been documented?”
   Ans: The quality manager asked the technician **whether the insulation resistance test had been documented**.
2. The lead architect enquired, “Can the server cluster handle double the transaction throughput?”
   Ans: The lead architect enquired **whether the server cluster could handle double the transaction throughput**.
3. The field auditor asked the safety engineer, “Did you inspect the gas leakage relief lines?”
   Ans: The field auditor asked the safety engineer **whether he/she had inspected the gas leakage relief lines**.
4. The client asked the team, “Will the firmware update fix the intermittent battery drain?”
   Ans: The client asked the team **whether the firmware update would fix the intermittent battery drain**.
5. The instructor asked the lab group, “Why did the multimeter show fluctuating voltage readings?”
   Ans: The instructor asked the lab group **why the multimeter had shown fluctuating voltage readings**.

Exercise 14: Countable and Uncountable Nouns

Identify the incorrect noun form and rewrite each sentence correctly.

1. The procurement officer purchased several new machineries and equipments for the workshop.
   Ans: Incorrect: *machineries, equipments*. Correct: The procurement officer purchased **new machinery and equipment** for the workshop (or “several new pieces of machinery and equipment”).
2. The laboratory generated a huge amount of test datas during the wind tunnel trials.
   Ans: Incorrect: *datas*. Correct: The laboratory generated a huge amount of test **data** during the wind tunnel trials.
3. We need three more softwares licenses to complete the finite element analysis on the model.
   Ans: Incorrect: *softwares*. Correct: We need three more **software** licenses to complete the finite element analysis on the model.
4. The field report provided numerous valuable informations regarding the soil composition.
   Ans: Incorrect: *informations*. Correct: The field report provided **a great deal of valuable information** (or “numerous valuable pieces of information”) regarding the soil composition.
5. All chemical apparatuses and glasswares must be washed thoroughly after the experiment.
   Ans: Incorrect: *glasswares* (and the uncountable usage of apparatuses). Correct: All chemical **apparatus and glassware** must be washed thoroughly after the experiment.

Exercise 15: Comparative Adjectives (Irregular Forms)

Fill in the blanks using the correct irregular comparative form.

1. Because the heat sink had poor thermal paste contact, heat dissipation was _____ (bad) than expected.
   Ans: worse

2. The redesigned microstrip antenna achieved a _____ (good) signal-to-noise ratio than the baseline patch.
   Ans: better

3. The autonomous vehicle traveled 50 km _____ (far) on a single charge after the weight reduction overhaul.
   Ans: farther / further

4. Switching to asynchronous task queues resulted in _____ (little) CPU idle time during peak batch processing.
   Ans: less

5. We require _____ (further / father) metallurgical investigation into why the stress cracks formed along the weld seam.
   Ans: further ("father" is not a comparative).

PART B

Exercise 1: Tenses (Past Tense Conversion & Industrial Reporting)

Rewrite the following descriptions in the Simple Past Tense.

1. "The technician inspects the solar arrays and records the peak power conversion value."
   Ans: The technician **inspected** the solar arrays and **recorded** the peak power conversion value.
2. "The chemical analyzer detects trace mercury levels and sends an automated alert to the environmental dashboard."
   Ans: The chemical analyzer **detected** trace mercury levels and **sent** an automated alert to the environmental dashboard.
3. "Maintenance crews lubricate the heavy conveyor bearings and replace the worn rubber belts before the morning shift."
   Ans: Maintenance crews **lubricated** the heavy conveyor bearings and **replaced** the worn rubber belts before the morning shift.
4. "The supervisory software polls field sensors every five milliseconds and archives temperature trends into a secure database."
   Ans: The supervisory software **polled** field sensors every five milliseconds and **archived** temperature trends into a secure database.
5. "The testing team applies a 500-volt dielectric pulse across the stator winding and monitors for insulation breakdown."
   Ans: The testing team **applied** a 500-volt dielectric pulse across the stator winding and **monitored** for insulation breakdown.

Exercise 2: Comparative Adjectives (Technical Metrics & Trade-Offs)

Draft two technical comparative sentences for each prompt.

1. Cloud Storage vs. On-Premise Physical Servers (initial deployment cost, elastic scalability).
   Ans: Cloud storage has a **lower** initial deployment cost than on-premise physical servers because no hardware must be purchased upfront. It is also **more elastic** than physical servers, since capacity can be scaled up or down within minutes.
2. Fiber-Optic vs. Copper Twisted-Pair Cables (bandwidth, EMI susceptibility).
   Ans: Fiber-optic cables offer a **much higher** data transfer bandwidth than copper twisted-pair cables. They are also **far less susceptible** to electromagnetic interference because they transmit light instead of electrical signals.
3. Electric Vehicles vs. ICE Vehicles (tailpipe emissions, purchase price).
   Ans: Electric vehicles produce **lower** tailpipe emissions than ICE vehicles, since they release none at all. However, EVs generally have a **higher** initial purchase price than comparable ICE vehicles.
4. Hydraulic vs. Electromechanical Actuators (force output, fluid leakage risk).
   Ans: Hydraulic actuators deliver a **greater** raw force output than electromechanical actuators. However, they carry a **higher** risk of fluid leakage, whereas electromechanical actuators are cleaner and safer to maintain.
5. Relational (SQL) vs. Document (NoSQL) Databases (schema rigidity, horizontal query throughput).
   Ans: Relational databases have a **more rigid** schema than document databases, which allow flexible and evolving data structures. NoSQL databases typically achieve a **higher** horizontal query throughput than relational databases because data can be distributed across many nodes more easily.

Exercise 3: Relative Clauses (Sentence Synthesis)

Combine each pair into a single sentence using a defining or non-defining relative clause.

1. (a) The algorithm analyzes microscopic cracks in turbine blades. (b) It was patented by the aeronautics team.
   Ans: The algorithm **that** analyzes microscopic cracks in turbine blades was patented by the aeronautics team.
2. (a) The lead chemical engineer authorized the emergency boiler shutdown. (b) She noticed an abnormal pressure spike in chamber two.
   Ans: The lead chemical engineer, **who** noticed an abnormal pressure spike in chamber two, authorized the emergency boiler shutdown.
3. (a) The laboratory houses a 400 kV impulse voltage generator. (b) High-voltage insulator puncture tests are conducted there.
   Ans: The laboratory, **where** high-voltage insulator puncture tests are conducted, houses a 400 kV impulse voltage generator.
4. (a) We engaged an external cybersecurity consultant. (b) His firmware vulnerability audit exposed three backdoor ports.
   Ans: We engaged an external cybersecurity consultant **whose** firmware vulnerability audit exposed three backdoor ports.
5. (a) The supervisory system utilizes an artificial neural network. (b) It predicts transformer insulation life based on dissolved gas concentrations.
   Ans: The supervisory system utilizes an artificial neural network **that** predicts transformer insulation life based on dissolved gas concentrations.

Exercise 4: If-Clauses (Technical Conditionals & Troubleshooting)

Complete the conditional statements logically and grammatically (model answers).

1. 1(a) If the operating temperature exceeds 85°C, ____
   Ans: …the cooling system **will switch** to high-speed mode and an alarm **will sound**.
2. 1(b) If the operator had pressed the emergency shutdown switch immediately, ____
   Ans: …the machine **would have stopped** before the damage **occurred**.
3. 2(a) If the supply voltage drops below the 200V threshold, ____
   Ans: …the under-voltage relay **will disconnect** the load to protect the motor.
4. 2(b) If the network administrator had enabled multi-factor authentication, ____
   Ans: …the attacker **would not have gained** access to the server.
5. 3(a) If toxic carbon monoxide concentrations surpass 50 ppm in the tunnel, ____
   Ans: …the exhaust fans **will start** automatically and the evacuation alarm **will be triggered**.
6. 3(b) If the mechanical team had checked the bearing lubrication schedule, ____
   Ans: …the bearing **would not have overheated** and failed.
7. 4(a) If the autonomous guided vehicle (AGV) detects an obstacle within 1 meter, ____
   Ans: …it **will stop** immediately and **reroute** along a clear path.
8. 4(b) If the thermal sensor had registered the runaway reaction in time, ____
   Ans: …the control system **would have shut down** the reactor and prevented the explosion.
9. 5(a) If the software receives an unauthenticated API token, ____
   Ans: …it **will reject** the request and **return** an authentication error.
10. 5(b) If the civil inspectors had reviewed the core sample density tests, ____
    Ans: …they **would have identified** the weak concrete before approving the structure.

Exercise 5: Impersonal Passive Voice

Convert the active sentences into the Impersonal Passive Voice for formal engineering reports.

1. "The chemical engineers distilled the crude extract at a constant temperature of 60°C."
   Ans: The crude extract **was distilled** at a constant temperature of 60°C.
2. "The metallurgist polished the cross-section of the weld seam with fine diamond paste."
   Ans: The cross-section of the weld seam **was polished** with fine diamond paste.
3. "The quality control division rejected fifteen cast-iron pump casings due to internal porosities."
   Ans: Fifteen cast-iron pump casings **were rejected** due to internal porosities.
4. "The technician measured the resistance across the bridge circuit using a calibrated digital ohmmeter."
   Ans: The resistance across the bridge circuit **was measured** using a calibrated digital ohmmeter.
5. "The flight test crew initiated the emergency aerodynamic braking flaps at an altitude of 3,000 meters."
   Ans: The emergency aerodynamic braking flaps **were initiated** at an altitude of 3,000 meters.

Exercise 6: Reported Speech (Interrogative Technical Transformations)

Transform the direct queries into Reported (Indirect) Speech.

1. The safety supervisor asked the intern, "Did you verify the grounding connections on the generator?"
   Ans: The safety supervisor asked the intern **whether he/she had verified the grounding connections on the generator**.
2. The plant director asked the site engineer, "Has the emergency effluent treatment plant received clearance?"
   Ans: The plant director asked the site engineer **whether the emergency effluent treatment plant had received clearance**.
3. The IT security lead enquired of the developer, "Why did you disable the firewall rule on the database server?"
   Ans: The IT security lead enquired of the developer **why he/she had disabled the firewall rule on the database server**.
4. The client asked the project manager, "Can the bridge structure withstand Category 4 hurricane winds?"
   Ans: The client asked the project manager **whether the bridge structure could withstand Category 4 hurricane winds**.
5. The research guide asked the postgraduate scholar, "How many sample trials did you conduct under cryogenic conditions?"
   Ans: The research guide asked the postgraduate scholar **how many sample trials he/she had conducted under cryogenic conditions**.

Exercise 7: Countable vs. Uncountable Nouns

Identify and correct the errors involving mass/uncountable nouns.

1. 1(a) We need to install several new software and machineries in the CAD laboratory.
   Ans: Error: *several new software, machineries*. Correct: We need to install **new software and machinery** (or "several new software packages and pieces of machinery") in the CAD laboratory.
2. 1(b) The team collected plenty of useful informations from the field test.
   Ans: Error: *informations*. Correct: The team collected plenty of useful **information** from the field test.
3. 2(a) The technician ordered ten new apparatuses and modern electronic equipments for the antenna lab.
   Ans: Error: *equipments* (and *apparatuses* as a mass noun). Correct: The technician ordered ten new **pieces of apparatus** and modern electronic **equipment** for the antenna lab.
4. 2(b) High-speed train operations require advanced transport infrastructures and dedicated electrical powers.
   Ans: Error: *infrastructures, powers*. Correct: High-speed train operations require advanced transport **infrastructure** and dedicated electrical **power**.

5. 3(a) The environmental sensor gathers extensive atmospheric datas and pollutions indices every minute.
   Ans: Error: *datas, pollutions*. Correct: The environmental sensor gathers extensive atmospheric **data** and **pollution** indices every minute.
6. 3(b) The construction crew unloaded tons of sands, cements, and steel debris at the worksite.
   Ans: Error: *sands, cements*. Correct: The construction crew unloaded tons of **sand, cement,** and steel debris at the worksite.
7. 4(a) All electrical hardwares and wirings must comply with local safety legislations.
   Ans: Error: *hardwares, wirings, legislations*. Correct: All electrical **hardware and wiring** must comply with local safety **legislation**.
8. 4(b) The data scientist discovered several interesting knowledges after running the machine learning clustering.
   Ans: Error: *knowledges*. Correct: The data scientist discovered several interesting **insights** (or "a great deal of new knowledge") after running the machine learning clustering.
9. 5(a) Laboratory staffs must clear away all glasswares and acidic wastes before leaving the premises.
   Ans: Error: *staffs, glasswares, wastes*. Correct: Laboratory **staff** must clear away all **glassware** and acidic **waste** before leaving the premises.
10. 5(b) The project failed due to a lack of proper fundings and technical researches.
    Ans: Error: *fundings, researches*. Correct: The project failed due to a lack of proper **funding** and technical **research**.

Exercise 8: Infinitives of Purpose & Gerund Constructions

Rewrite the pairs using an infinitive of purpose (to + verb) or a gerund after a preposition (by / for + -ing).

1. 1(a) A pressure relief valve is installed. It vents excess boiler steam.
   Ans: A pressure relief valve is installed **to vent** excess boiler steam.
2. 1(b) The automated script runs every midnight. It archives legacy server logs.
   Ans: The automated script runs every midnight **to archive** legacy server logs.
3. 2(a) Automotive engineers installed an intercooler. It reduces the compressed air intake temperature.
   Ans: Automotive engineers installed an intercooler **to reduce** the compressed air intake temperature.
4. 2(b) The technicians analyzed the structural stress points. They placed piezoelectric strain gauges along the girder.
   Ans: The technicians analyzed the structural stress points **by placing** piezoelectric strain gauges along the girder.
5. 3(a) High-performance heatsinks feature copper heatpipes. They conduct heat away from the silicon die.
   Ans: High-performance heatsinks feature copper heatpipes **to conduct** heat away from the silicon die.
6. 3(b) The factory cut water consumption by 30%. They implemented a closed-loop recycling filtration system.
   Ans: The factory cut water consumption by 30% **by implementing** a closed-loop recycling filtration system.
7. 4(a) Safety guards are mounted over rotating flywheel belts. They prevent accidental worker contact.
   Ans: Safety guards are mounted over rotating flywheel belts **to prevent** accidental worker contact.
8. 4(b) The algorithm optimizes memory bandwidth. It pre-fetches sequential array indices into L2 cache.
   Ans: The algorithm optimizes memory bandwidth **by pre-fetching** sequential array indices into L2 cache.
9. 5(a) The electronics lab installed an uninterruptible power supply (UPS). It protects servers against sudden voltage sags.
   Ans: The electronics lab installed an uninterruptible power supply (UPS) **to protect** servers against sudden voltage sags.

10. 5(b) The metallurgy group enhanced surface hardness. They diffused nitrogen atoms into the steel casing.
    Ans: The metallurgy group enhanced surface hardness **by diffusing** nitrogen atoms into the steel casing.

Exercise 9: Adverbs of Manner & Operational Precision

Fill in the blanks with the appropriate adverbial forms.

1. 1(a) The drone adjusted its flight path _____ (automatic) to avoid overhead power cables.
   Ans: automatically
2. 1(b) The software patch was integrated _____ (flawless) into the production server.
   Ans: flawlessly
3. 2(a) The CNC cutting head traced the complex automotive profile _____ (accurate).
   Ans: accurately
4. 2(b) The turbine governor responded _____ (instant) to the unexpected grid frequency drop.
   Ans: instantly
5. 3(a) The hazardous gas extraction vents operate _____ (continuous) throughout the electroplating process.
   Ans: continuously
6. 3(b) Technicians handled the radioactive isotope capsule _____ (cautious) using robotic arm manipulators.
   Ans: cautiously
7. 4(a) The optical pyrometer recorded the furnace temperature _____ (consistent) across three production shifts.
   Ans: consistently
8. 4(b) The structural frame buckled _____ (catastrophic) under the simulated earthquake test load.
   Ans: catastrophically
9. 5(a) The firmware routine checks for incoming serial packets _____ (periodic) every hundred microseconds.
   Ans: periodically
10. 5(b) The titanium alloy workpiece was machined _____ (smooth) to a mirror finish.
    Ans: smoothly

Exercise 10: Modals of Capability, Obligation, and Prohibition

Insert an appropriate modal auxiliary conveying the meaning in brackets.

1. 1(a) The microcontroller _____ (capability) drive up to six stepper motors simultaneously.
   Ans: can
2. 1(b) Laboratory staff _____ (strict prohibition) handle corrosive reagents without full face shields.
   Ans: must not (also: may not / cannot)
3. 2(a) Before entering the cleanroom chamber, all personnel _____ (mandatory obligation) put on antistatic footwear.
   Ans: must (also: have to)
4. 2(b) Given the high ambient humidity, water droplet condensation _____ (moderate possibility) form inside the optical housing.
   Ans: may / might
5. 3(a) Unauthorized maintenance personnel _____ (strict prohibition) override the high-voltage interlock switch.
   Ans: must not (also: may not / cannot)

6. 3(b) The upgraded wireless transmitter _____ (past capability) broadcast telemetry signals up to forty kilometers.
   Ans: could

7. 4(a) Quality inspection teams _____ (strong recommendation / advice) recalibrate measuring calipers every ninety days.
   Ans: should

8. 4(b) The digital ammeter is reading zero amperes; the auxiliary fuse _____ (logical deduction / certainty) be blown.
   Ans: must

9. 5(a) Industrial robots _____ (capability) perform repetitive spot welds with sub-millimeter precision.
   Ans: can

10. 5(b) Visitors _____ (strict prohibition) enter the laser testing enclosure while the amber warning beacon is active.
    Ans: must not (also: may not / cannot)

Topic 1: Writing Imperative Safety Instructions (CO1 | BL3)

Target: 8 clear, numbered, imperative instructions using modals/action verbs.

Question 1: Write eight imperative safety instructions for a Chemistry & Materials Testing Laboratory to avoid chemical burns, inhalation hazards, and equipment contamination.

Model Answer

SAFETY INSTRUCTIONS – CHEMISTRY & MATERIALS TESTING LABORATORY

1. Wear a lab coat, splash-proof goggles, nitrile gloves, and closed-toe shoes at all times.
2. Handle volatile, toxic, or concentrated chemicals only inside a working fume hood; never smell or taste any substance.
3. Always add acid to water slowly while stirring; never add water to acid.
4. Label every container clearly and never return unused chemicals to the stock bottles.
5. Do not pipette by mouth; use a pipette filler or a dispenser.
6. Clean all glassware and testing instruments before and after use, and wipe up spills immediately to prevent contamination.
7. Dispose of chemical waste only in the designated, labelled containers; never pour it down the sink.
8. Locate the eyewash station, safety shower, and fire extinguisher before starting work, and report every spill or burn to the instructor at once.

Question 2: Write eight imperative safety instructions for a High-Voltage Electrical Machines & Drives Laboratory to prevent electric shock, arc flash hazards, and accidental short circuits during motor loading experiments.

Model Answer

SAFETY INSTRUCTIONS – ELECTRICAL MACHINES & DRIVES LABORATORY

1. Do not switch on the supply until the instructor has inspected and approved your circuit.
2. Wear insulated footwear and safety glasses, and remove all metallic jewellery and watches before starting.
3. Make or change every connection only when the supply is switched off and isolated.
4. Use only properly rated, insulated leads and check them for damage before use.
5. Ensure that the machine frame is earthed and that all couplings and rotating parts are guarded before loading.
6. Start the motor at the minimum starter/rheostat setting, increase the load gradually, and never exceed the rated current or voltage.
7. Never touch live terminals or rotating parts, keep the floor dry, and never work alone.
8. In an emergency, press the emergency stop, switch off the main supply, and use a CO_2 extinguisher (never water) on electrical fires.

Question 3: Write eight imperative safety guidelines for an Advanced Manufacturing and CNC Machining Workshop to prevent injuries from high-speed spindle operations, hot metallic chip expulsion, coolant spills, and moving cutter heads.

Model Answer

SAFETY GUIDELINES – ADVANCED MANUFACTURING & CNC MACHINING WORKSHOP

1. Wear safety glasses, safety shoes, and close-fitting clothes; tie back long hair and remove rings, watches, and loose gloves.
2. Close the guard door before starting the spindle, and never bypass the safety interlocks.
3. Clamp the workpiece and tool firmly, and remove chuck keys and wrenches before pressing the start button.
4. Keep your hands away from the spindle and cutter head, and wait until all motion stops before measuring or adjusting.
5. Never remove hot chips by hand; use a brush or chip hook after the machine stops, and wear a face shield when chips are flying.
6. Wipe up coolant spills immediately and keep the floor dry and free of oil to prevent slipping.
7. Verify the program and tool offsets with a dry run at low feed rate before cutting.
8. Know the location of the emergency stop button, never leave a running machine unattended, and report any malfunction or injury to the supervisor immediately.

Topic 2: Free Writing – Festivals & Cultural Celebrations (CO2 | BL4)

Target: Descriptive essay of 250 words integrating community customs, festive meals, and sustainable practices.

Question 1: Write a descriptive essay (250 words) on "A Traditional Harvest Festival in Your Region and How Modern Eco-Friendly Practices Can Preserve Its Core Spirit."

Model Answer

Pongal: A Harvest Festival in Harmony with Nature

Pongal, the harvest festival of Tamil Nadu, is celebrated every January to thank the Sun, the earth, and the cattle for a bountiful harvest. The festival lasts four days: Bhogi, Thai Pongal, Mattu Pongal, and Kaanum Pongal. Villages wake before dawn, houses are cleaned, and colourful kolams drawn with rice flour decorate every doorstep.

The heart of the celebration is the cooking of fresh rice with milk and jaggery in a new clay pot in the open courtyard. When the milk boils over, families shout "Pongalo Pongal!" and offer the dish to the Sun. The festive meal includes sweet pongal, venn pongal, sugarcane, bananas, and sambar served on banana leaves. On Mattu Pongal, cattle are bathed, painted, and garlanded, while neighbours gather for folk songs, kabaddi, and village games that strengthen community bonds.

Modern life, however, has added plastic, smoke, and waste to the celebration. Eco-friendly practices can preserve its core spirit. Clay pots, banana leaves, and areca-leaf plates can replace plastic and thermocol. Kolams made of rice flour feed ants and birds. Instead of burning old tyres and plastics in Bhogi bonfires, communities can collect old items for recycling and compost dry leaves. Community kitchens can use biogas or efficient stoves, and leftover food can be composted or shared with the needy. Decorations of mango leaves, flowers, and sugarcane are fully biodegradable.

By blending tradition with sustainability, Pongal can continue to honour nature while protecting it. A festival that celebrates the earth must, above all, take care of it.

Question 2: Write a descriptive essay (250 words) on "A Traditional Temple Car or Float Festival in Your Hometown and Sustainable Measures to Minimize Single-Use Plastics and Noise Pollution."

Model Answer

The Temple Car Festival: Faith with Responsibility

The annual temple car (Therottam) festival in my hometown is the most awaited event of the year. Every spring, thousands of devotees gather outside the ancient temple to pull the massive wooden chariot carrying the festival deity through the four mada streets. Days before the event, the temple is whitewashed, the streets are decorated with kolams and banana trees, and the town resonates with nadaswaram music and chanting.

On the main day, the idol is adorned with silk and jasmine garlands and placed on the car. Men, women, and children hold the thick ropes and pull together, chanting in unison, which symbolises unity and collective devotion. Along the route, families offer coconuts, camphor, and fruits, while volunteers distribute prasadam such as pongal, puliyodarai, sundal, and panakam on areca-leaf plates. Small fairs sell toys, bangles, and sweets.

Unfortunately, the festival also generates plastic cups, packets, and loudspeaker noise. Several measures can reduce this impact. The temple trust can ban single-use plastics and provide steel tumblers, banana leaves, and refill-based drinking water points. Volunteer teams can segregate waste at collection points and compost food scraps. Noise can be controlled by limiting loudspeaker decibel levels, replacing firecrackers with traditional instruments, and keeping cultural programmes within fixed hours. Crowd management can be improved with barricaded lanes, trained volunteers, one-way movement, and clearly marked medical and water points.

By combining faith with responsibility, the Therottam can retain its spiritual grandeur while keeping the town clean, peaceful, and safe for future generations.

Question 3: Write a descriptive essay (250 words) on "Celebrating Regional Folk Art and Handicraft Fairs Using Renewable Solar Micro-Grids and Zero-Waste Composting."

Model Answer

Folk Art Fairs: Heritage Powered by the Sun

Every winter, our town hosts a regional folk art and handicraft fair where the streets become a living museum of heritage. Stalls display Thanjavur paintings, Kalamkari fabrics, terracotta figures, palm-leaf baskets, and brass lamps. Artisans demonstrate weaving, pottery, and wood carving, while folk groups perform Karagattam, Oyilattam, and Therukoothu in the evenings. The air is filled with drumbeats, the smell of earthen lamps, and the laughter of children, creating a warm and festive atmosphere.

Food is equally central to the experience. Visitors queue for heritage delicacies such as kambu koozh, ragi kali, kuzhi paniyaram, palm-jaggery sweets, murukku, and filter coffee served in steel tumblers. Millets and local produce are promoted in place of industrial snacks.

Modern environmental strategies can make such fairs sustainable. A renewable solar micro-grid, built with rooftop panels and battery storage, can power the stalls, LED lights, and sound systems without noisy diesel generators, cutting emissions and noise. Surplus energy can even be shared with nearby homes. Zero-waste composting is equally important: food stalls can use leaf plates and clay cups, and all leftovers can be collected in colour-coded bins and sent to community compost pits that produce manure for local gardens. Dry waste can be recycled, and water refill stations can replace plastic bottles.

Such measures also educate visitors by showing that tradition and technology can work together. When artisans' skills are supported by clean energy and responsible waste management, folk culture is preserved while the environment is protected.

Topic 3: Free Writing – Food & Technical Travel Plans (CO2 | BL4)

Target: Structured travel narrative (250 words) combining route logistics, local dining experiences, and technical landscape observations.

Question 1: Write a structured travel plan and descriptive feature (250 words) on "A Technical Study Tour to a Major Hydroelectric Dam."

Model Answer

Technical Study Tour: Mettur Dam Hydroelectric Station

Route and logistics: Our group of forty second-year students will leave Chennai by a chartered bus at 5:00 a.m. and travel about 340 km via Vellore and Salem to Mettur, reaching the dam by early afternoon. The return journey will be the next evening. Accommodation will be in a nearby guest house, and prior permission from the dam authorities will be obtained for the technical visit.

Technical observations: Built across the Cauvery River in 1934, the Mettur Dam forms the vast Stanley Reservoir. From the dam crest, we will observe the masonry gravity structure, the 16-gate spillway, and the intake towers. Engineers will take us through the powerhouse, where water flowing through penstocks spins the turbines and generators. We will study the head, flow rate, governors, transformers, and the switchyard that transmits power to the grid. The surrounding hills and the calm reservoir offer a perfect view of how water management and power generation coexist.

Local food: After the visit, we will taste the region's famous Mettur fish fry beside the dam, along with Kongunadu-style meals of ragi kali, kambu koozh, and spicy country-chicken gravy. The next morning, we will enjoy Salem's famous mangoes and soft idlis with sambar.

Safety: Students must wear closed shoes, follow the guide's instructions, and avoid restricted zones.

This tour will connect classroom theory on fluid mechanics and power generation with real-world practice.

Question 2: Write a structured travel plan and descriptive feature (250 words) on "A Technical Study Tour to an Integrated Solar and Wind Hybrid Energy Farm."

Model Answer

Technical Study Tour: Wind–Solar Hybrid Energy Farm, Muppandal

Route and logistics: Our team of forty students will board an overnight train from Chennai Egmore to Nagercoil, a journey of about twelve hours. From there, a hired mini-bus will take us roughly 25 km to the wind–solar hybrid farm near Muppandal, one of India's oldest wind-power regions. Accommodation will be arranged at a budget lodge in Nagercoil, and entry permission will be taken in advance.

Technical observations: The Aralvaimozhi gap channels strong winds between the hills, so hundreds of turbines turn in steady rhythm across the landscape. We will study the tower, nacelle, gearbox, blade pitch control, and the generator. Beside them, rows of photovoltaic arrays convert sunlight directly into electricity, and inverters and transformers condition the power before it enters the grid. The key observation will be how wind peaks in the afternoon while solar peaks at noon, so the hybrid system delivers steadier output, supported by an energy management system.

Local food: Along the way, we will sample Nanjil Nadu specialities such as fish curry meals, adai with aviyal, and crisp banana chips, and we will carry Tirunelveli halwa home as a sweet memory.

Safety: Everyone must wear helmets and safety shoes, stay within marked areas, and avoid going near rotating blades or electrical yards.

This tour will convert our knowledge of renewable energy into practical understanding.

Question 3: Write a structured travel plan and descriptive feature (250 words) on "A Technical Field Visit to an Automated Seawater Reverse Osmosis (RO) Desalination Plant."

Model Answer

Field Visit: Seawater Reverse Osmosis Desalination Plant, Nemmeli

Route and logistics: Our group of forty students will leave the college campus at 8:00 a.m. in two hired buses and travel along the scenic East Coast Road to the Nemmeli desalination plant, about 35 km away. The visit is scheduled from 10:30 a.m. to 1:30 p.m., after prior written approval from the plant authorities. We will return by 5:00 p.m.

Technical observations: At the *intake pumping station*, seawater is drawn through submerged pipes and screens. We will follow it through pre-treatment filters that remove sand and suspended particles, and then to the long *membrane filtration racks*, where high-pressure pumps force water through semi-permeable membranes, leaving the salts behind. Energy recovery devices recapture pressure from the reject stream and cut power consumption. Finally, the *brine discharge outlets* release the concentrated salt stream through diffusers into the sea, where it is diluted to protect marine life. A SCADA control room monitors the pressure, flow, and conductivity automatically.

Coastal food experience: On the way back, we will stop at a seaside restaurant for fried fish, prawn masala, and crab curry with steamed rice, followed by tender coconut and hot filter coffee.

Safety: Students must wear closed shoes and helmets, keep away from high-pressure lines, and follow the guide at all times.

This visit will show how engineering can turn seawater into a reliable source of drinking water.

Topic 4: Formal Letters of Request (CO3 | BL4)

Target: Formal business letter layout with sender/receiver address, clear subject line, technical rationale, specific requirements, and courteous sign-off.

Question 1: Draft a formal Letter of Request to the Head of the Computer Science Department requesting permission and lab access after college hours for your project team to prepare for a National AI Hackathon.

Model Answer

[Your Name]
Department of Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Date: 04 November 2026

The Head of the Department
Department of Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Subject: Request for permission to use the computer laboratory after college hours for National AI Hackathon preparation

Respected Sir/Madam,

We, a team of four second-year students, have been shortlisted for the National AI Hackathon to be held on 20 and 21 December 2026. We are developing an AI-based system that detects crop diseases from leaf images using a convolutional neural network, with the aim of helping farmers receive quick and accurate advice.

To prepare effectively, our team needs to train and test the model on a high-performance system, which is not available in our hostels. We therefore request permission to use the Computer Science laboratory from 5:00 p.m. to 8:00 p.m. on weekdays from 10 November to 15 December 2026. We will require: (1) GPU-enabled workstations, (2) Python, TensorFlow, PyTorch, and Jupyter Notebook, (3) high-speed internet access to download datasets, and (4) access to the project server for version control.

We assure you that we will follow all laboratory rules, handle the equipment carefully, and switch off all systems before leaving. If required, a faculty member may supervise our sessions.

We kindly request you to grant us permission. We shall be grateful for your support.

Thanking you,

Yours faithfully,
[Your Name]
(On behalf of Team [Team Name])
Register No.: [____]

Question 2 (Industrial Visit): Write a formal Letter of Request to the Plant Manager of an Industrial Robotics & Automation Factory seeking approval for an industrial plant visit for 50 first-year engineering students, outlining the intended learning outcomes.

Model Answer

The Head, Department of Mechanical Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Date: 04 November 2026

The Plant Manager
[Name of Factory], Industrial Robotics & Automation Division
[Address of Factory]

Subject: Request for permission to conduct an industrial visit for first-year engineering students

Respected Sir/Madam,

I am writing on behalf of the Department of Mechanical Engineering, St. Joseph's College of Engineering, to request your kind permission for 50 first-year engineering students, accompanied by two faculty members, to visit your robotics and automation plant on any convenient weekday between 15 and 30 November 2026, from 10:00 a.m. to 1:00 p.m.

The visit aims to give our students exposure to modern industrial practice. The intended learning outcomes are: (1) to understand how robotic arms, PLCs, and sensors are used in automated assembly lines; (2) to observe quality-control and safety systems in a real production environment; (3) to relate classroom concepts of mechanics, electronics, and programming to industry; and (4) to learn about career opportunities in automation and robotics.

We will strictly follow your safety rules and dress code, and the students will remain under faculty supervision at all times. We would be grateful if you could also depute an engineer to brief the students.

We look forward to your favourable reply and thank you for your time and support.

Yours faithfully,
[Name]
Head of the Department
Contact: [phone / email]

Question 3 (Practice A – Workshop Facilities): Draft a formal Letter of Request to the Head of the Mechanical Engineering Department seeking permission and machine shop access after college hours for your student team to fabricate and test an energy-efficient electric go-kart for an inter-collegiate design challenge.

Model Answer

[Your Name]
Department of Mechanical Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Date: 04 November 2026

The Head of the Department
Department of Mechanical Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Subject: Request for machine shop access after college hours for fabrication of an electric go-kart

Respected Sir/Madam,

Our six-member student team has registered for the Inter-Collegiate Design Challenge to be held in January 2027. We plan to design, fabricate, and test an energy-efficient electric go-kart powered by a brushless DC motor and a lithium-ion battery pack, with a lightweight steel-tube chassis to maximize range.

Since regular class hours do not allow enough time for fabrication, we request permission to use the machine shop from 4:30 p.m. to 7:30 p.m. on weekdays and on Saturdays from 9:00 a.m. to 1:00 p.m., from 10 November 2026 to 10 January 2027. Our requirements are: (1) welding and cutting equipment, (2) lathe and drilling machines, (3) a workbench and measuring tools, and (4) space to assemble and test the vehicle.

We promise to wear proper safety gear, to operate the machines only under the technical staff's supervision, and to keep the workshop clean. We will also share our design reports with the department.

We shall be highly obliged if you would kindly grant us permission.

Thanking you,

Yours faithfully,
[Your Name]
Team Captain, Team [Name]
Register No.: [____]

Question 4 (Practice B – Sponsorship & Duty Leave): Draft a formal Letter of Request to the College Principal / Dean of Student Affairs requesting financial sponsorship and duty leave for a five-member multidisciplinary student team selected to present a drone surveillance prototype at a National Defense Innovation Expo.

Model Answer

[Your Name]
Department of [____]
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Date: 04 November 2026

The Principal / Dean of Student Affairs
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Subject: Request for financial sponsorship and duty leave to present a drone surveillance prototype at the National Defense Innovation Expo

Respected Sir/Madam,

We are happy to inform you that our five-member multidisciplinary team (Mechanical, Electronics, Computer Science, and Aeronautical students) has been selected to present its AI-assisted drone surveillance prototype at the National Defense Innovation Expo, to be held in New Delhi from 12 to 14 December 2026.

The prototype uses a quadcopter with a thermal camera and an onboard object-detection algorithm to monitor border-like terrain in real time. Presenting it at the Expo will give our college national visibility and help us receive expert feedback.

We therefore request (1) **duty leave** for all five members from 10 to 16 December 2026, including travel days, and (2) **financial sponsorship** of approximately Rs. 60,000, covering train fares, accommodation, registration fees, and transportation of the prototype. A detailed budget is attached.

We assure you that we will make up for missed classes and will submit a report after the Expo.

We shall be extremely grateful for your kind consideration and support.

Thanking you,

Yours faithfully,
[Your Name]
(On behalf of the team)
Enclosure: Selection letter and budget estimate

Topic 5: Compare & Contrast Descriptive Essays (CO4 | BL5)

Target: Analytical essay (250–300 words) using comparative transitions to evaluate technical trade-offs.

Question 1: Compare and contrast the positives and negatives of Electric Vehicles (EVs) versus Conventional Internal Combustion Engine (ICE) Vehicles.

Model Answer

Electric Vehicles versus Internal Combustion Engine Vehicles

Electric vehicles and conventional ICE vehicles represent two very different approaches to personal transport, each with clear strengths and weaknesses.

In terms of battery life and fuel, an EV battery typically lasts eight to ten years before its capacity falls noticeably, and replacement is costly. An ICE vehicle, on the other hand, relies on an engine that can last far longer with regular servicing, although it needs constant fuel. Regarding the manufacturing footprint, EVs have a higher production impact because mining lithium, cobalt, and nickel for batteries is energy-intensive. In contrast, ICE vehicles have a lower manufacturing footprint but produce continuous tailpipe emissions throughout their life, whereas EVs emit none while driving.

Operating costs favour EVs. Electricity is cheaper per kilometre than petrol or diesel, and EVs have fewer moving parts, so servicing costs less. Conversely, ICE vehicles need oil changes, filters, and more frequent repairs, and they are exposed to fluctuating fuel prices. However, the purchase price of an EV is still higher than that of a comparable ICE vehicle.

Charging network feasibility remains the biggest challenge for EVs. Although charging stations are growing in cities, they are scarce on highways and in small towns, and charging takes much longer than refuelling. ICE vehicles, by contrast, enjoy a mature and widespread fuel network and refill within minutes.

In conclusion, EVs are better for the environment and cheaper to run, whereas ICE vehicles are more convenient and cheaper to buy today. As charging networks expand and battery costs fall, the balance is likely to shift towards electric mobility.

Question 2: Compose an analytical essay comparing the positives and negatives of Remote/Virtual Internships versus On-Site Physical Internships for early-stage engineering students.

Model Answer

Remote Internships versus On-Site Internships

For early-stage engineering students, internships provide the first glimpse of professional life. Remote and on-site internships both offer value, but in different ways.

Regarding hands-on skill acquisition, on-site internships are clearly superior. Students can handle real machines, instruments, and prototypes, and they learn workshop discipline. Remote internships, in contrast, depend mainly on simulations, software, and coding tasks, so they suit computing and design roles but limit practical exposure in mechanical, electrical, or civil fields.

In terms of schedule flexibility, remote internships have a distinct advantage. Students can balance them with classes, avoid travel and accommodation costs, and even work for companies in other cities. On-site internships, on the other hand, require fixed hours, relocation, and higher expenses, which may disrupt academic schedules.

When it comes to mentorship, on-site interns benefit from direct supervision, spontaneous questions, and immediate feedback. Remote mentoring, whereas, often relies on scheduled video calls, which can leave students feeling isolated or reluctant to ask doubts.

Professional networking follows a similar pattern. Physical presence helps students build relationships with colleagues, observe workplace culture, and receive recommendations. Virtual interns conversely meet fewer people, and their connections tend to remain limited.

In conclusion, on-site internships are more effective for practical skills, mentorship, and networking, whereas remote internships are more flexible and economical. A hybrid model that combines both may offer first-year engineers the best balance of learning and convenience.

Question 3: Write an analytical Compare and Contrast essay (250–300 words) evaluating the positives and negatives of Cloud Computing Infrastructure versus On-Premise Physical Data Centers for engineering enterprises.

Model Answer

Cloud Computing versus On-Premise Data Centers

Engineering enterprises must decide whether to host their applications and data on the cloud or in their own physical data centres. Each option carries distinct advantages and drawbacks.

With respect to upfront capital expenditure, the cloud is clearly attractive. Companies pay only for the resources they use, so they avoid buying servers, cooling systems, and real estate. On-premise data centres, in contrast, demand a large initial investment before a single application runs.

Recurring overheads, however, tell a different story. Cloud subscription fees can accumulate quickly for heavy, continuous workloads, and unexpected usage can inflate bills. On the other hand, once an on-premise facility is built, its running costs are more predictable, although the enterprise still pays for electricity, maintenance, and skilled staff.

Data security and compliance are two sides of the same coin. Leading cloud providers invest heavily in encryption and certified security, but sensitive design files or defence-related data may be subject to rules requiring local storage. On-premise systems give the enterprise full control over its data; conversely, they place the entire burden of protection on the company’s own staff.

Workload scalability is the cloud’s greatest strength. Computing power for simulations or testing can be increased within minutes and reduced afterwards. Physical data centres, whereas, are limited by installed hardware, and expanding them takes weeks or months.

In summary, the cloud offers flexibility and low initial cost, whereas on-premise data centres offer control and predictable long-term expenses. Many enterprises now adopt a hybrid approach to enjoy the benefits of both.

Question 4: Write a balanced Compare and Contrast essay (250–300 words) discussing the positives and negatives of 3D Printing (Additive Manufacturing) versus Traditional CNC Milling (Subtractive Manufacturing) in prototype fabrication.

Model Answer

3D Printing versus CNC Milling in Prototype Fabrication

Additive manufacturing (3D printing) builds parts layer by layer, whereas subtractive CNC milling carves them from a solid block. Both are widely used for prototyping, and each has its own trade-offs.

In terms of material wastage, 3D printing has a distinct advantage because it uses only the material needed for the part and its supports. CNC milling, in contrast, removes up to 90 percent of the block as chips, although this metal can often be recycled.

Regarding production speed for batch runs, the picture reverses. A 3D printer may take many hours to complete a single part, so printing large batches is slow. On the other hand, once a CNC machine is programmed, it can produce many identical parts quickly and consistently, making it more efficient for batch production.

Surface finish and accuracy favour CNC milling. It achieves tight tolerances and smooth surfaces directly from the machine, and it works with strong engineering metals. 3D-printed parts, conversely, often show visible layer lines and may need sanding or post-processing to meet precision requirements.

Geometric complexity is where 3D printing shines. It can create internal channels, lattice structures, and organic shapes that are impossible or very expensive to machine. CNC milling, however, is limited by tool access, so deep cavities and undercuts require special setups or multiple operations.

In conclusion, 3D printing is ideal for complex, low-volume prototypes with minimal waste, whereas CNC milling is better for accurate, smooth, and repeatable parts. Choosing between them depends on the design complexity, quantity, and finish required.

Topic 6: Personal Email Writing (CO5 | BL6)

Target: Personal email (informal yet polite and articulate tone) conveying reflections, academic transitions, and structured study plans.

Question 1: Draft a personal email to your high school mentor or teacher sharing your first-semester experiences at engineering college.

Model Answer

To: mentor.name@example.com
Subject: My First Semester at Engineering College – Thank You for Your Guidance

Dear Sir/Madam,

I hope you are doing well. I am writing to share my first-semester experiences at St. Joseph's College of Engineering and to thank you for the guidance that helped me reach here.

The biggest change has been the transition to independent campus life. Living in the hostel has taught me to manage my own time, laundry, and budget, and I have made good friends from different states. At first, I missed home, but a regular routine and weekly calls to my parents have helped.

Academically, I have enjoyed the laboratory sessions the most. In the Basic Electrical lab, I built my first series–parallel circuit, and in the programming lab, I wrote a small C program to calculate the average marks of a class. Working in teams has taught me patience and the value of recording observations carefully.

My end-semester examinations begin next month, so I have prepared a study plan. I revise two subjects a day, solve previous question papers on weekends, and practise numerical problems every evening. I have also formed a small study group to clear doubts.

Your advice to "stay curious and consistent" has stayed with me, and I would be grateful for any further suggestions you have for me. I would love to visit the school during the holidays.

With warm regards,
[Your Name]

Question 2: Draft a personal email to your high school science teacher reflecting on your first term in engineering, focusing on hands-on lab coursework, your first technical club or hackathon, and your strategy for the upcoming university theory and practical exams.

Model Answer

To: science.teacher@example.com
Subject: Reflections on My First Term in Engineering

Dear Ma'am/Sir,

I hope you and everyone at school are well. It has been almost four months since I joined engineering, and I wanted to share how things have been going, since your science classes inspired my choice.

Adapting to hands-on lab coursework was exciting but challenging. Unlike school experiments, we now work with oscilloscopes, breadboards, and microcontrollers, and a small wiring error can ruin the whole circuit. I have learned to read the manual first, take careful readings, and write neat observations, which you always insisted on.

I also joined the college's Robotics Club and, last month, took part in my first hackathon. Our team built a simple line-following robot in 24 hours. We did not win, but we learned about teamwork, debugging, and handling pressure.

Now I am preparing for the university theory and practical examinations. My strategy is to divide each syllabus into units, revise one unit every day, and solve past question papers. For practicals, I am repeating every experiment and revising the viva questions with my lab partner.

I would be glad to hear any advice you may have. Please convey my regards to all the teachers.

Yours sincerely,
[Your Name]

Question 3: Draft a personal email to your elder sibling or college alumnus seeking advice on balancing academic requirements with co-curricular technical clubs, describing your daily schedule, challenges in managing lab record submissions, and asking for tips on semester exam revision.

Model Answer

To: elder.sibling@example.com
Subject: Need Your Advice on Balancing Studies and Clubs

Hi Anna/Akka,

How are you? I hope work and life are going well. I am writing because I badly need your advice, since you handled college life so well.

As you know, I recently joined the IEEE student chapter and the coding club. My day starts at 6:30 a.m.; classes run from 9:00 to 4:30, and the club meets from 5:00 to 6:30 p.m. After dinner, I study for two hours or work on projects. I enjoy everything, but I often feel tired and short of time.

My biggest challenge is managing lab record submissions. Each experiment needs a neat record with circuit diagrams, observations, and results, and I usually postpone writing until the deadline, which creates last-minute stress. Sometimes I also forget the viva questions.

Could you please tell me how you balanced clubs and studies? Should I reduce my club hours during exam time? Do you have any tips for writing records regularly, perhaps by finishing them the same day?

Also, my end-semester exams are in a month, and I would love your tips for revision, such as how to make short notes, how many previous papers to solve, and how to handle difficult subjects like Engineering Mathematics.

Thanks a lot in advance. Please give my regards to everyone there, and let’s talk this weekend.

Love,
[Your Name]

Topic 7: Travel Planning & Budgetary Itineraries (CO5 | BL6)

Target: Organized travel schedule and budget breakdown (250 words) with day-by-day technical activities and safety measures.

Question 1: Write a Travel Plan and Budgetary Itinerary (250 words) for a 4-day student technical tour to an Aerospace and Satellite Launch Station.

Model Answer

4-Day Technical Tour: Satish Dhawan Space Centre (Sriharikota)

Participants and transport: 40 students and 2 faculty members travel in a chartered AC bus from Chennai (about 100 km), after obtaining official visitor permission and ID verification.

Day 1: Depart at 6:00 a.m., reach Sullurpeta by 9:00 a.m., and check in at a lodge. Afternoon briefing on rocket propulsion and launch operations.
Day 2: Visit the Space Centre: launch pads, the Vehicle Assembly Building, and the launch viewing gallery; interact with scientists about satellite launch vehicles and telemetry.
Day 3: Return to Chennai and visit Birla Planetarium and a science centre to study orbital mechanics and satellite applications.
Day 4: Group discussion, project presentations, and preparation of the technical report.

Budget (per student):
Bus hire (share): Rs. 1,200
Accommodation (2 nights): Rs. 1,600
Food: Rs. 1,200
Entry and permits: Rs. 300
Insurance and contingency: Rs. 300
Total: Rs. 4,600 (about Rs. 1,84,000 for 40 students)

Safety protocols: Carry photo ID; follow security instructions strictly; do not photograph restricted areas; wear closed shoes and stay with the group; maintain a first-aid kit and emergency contact list; and obey the guide at all times.

Question 2: Write a Travel Plan and Budgetary Itinerary (250 words) for a 4-day student industrial exposure tour to a Major Commercial Port and Maritime Dry-Dock Facility.

Model Answer

4-Day Industrial Exposure Tour: Cochin Port and Shipyard, Kochi

Transport arrangements: 40 students and 2 faculty members travel by reserved sleeper-class train from Chennai to Ernakulam (about 12 hours) and use a hired mini-bus for local movement.

Day 1: Overnight journey; arrive next morning, check in at a hostel, and rest.
Day 2: Technical briefing at the port: container terminal, cranes, berthing, and cargo handling systems.
Day 3: Visit to the shipyard's dry dock: ship repair, hull inspection, welding, and propulsion systems; briefing on naval architecture and dock operations.
Day 4: Interaction with port engineers, a short harbour cruise, and an evening return journey.

Budget (per student):
Train fare (return): Rs. 1,600
Stay (3 nights): Rs. 2,400
Food: Rs. 2,000
Local transport: Rs. 800
Entry fees and safety kits: Rs. 400
Contingency: Rs. 300
Total: Rs. 7,500 (about Rs. 3,00,000 for 40 students)

Dockyard safety protocols: Wear helmets, safety shoes, and reflective jackets; stay inside marked walkways; do not go near cranes, edges of the dock, or welding areas; never use phones near hazardous zones; carry life jackets on the harbour cruise; and follow the guide's instructions.

Question 3: Write a Travel Plan and Budgetary Itinerary (250 words) for a 4-day technical study tour to an Electric Locomotive Manufacturing Works and Automated Railway Signalling Center.

Model Answer

4-Day Technical Study Tour: Chittaranjan Locomotive Works and Railway Signalling Centre

Rail transit bookings: 40 students and 2 faculty members assemble at Howrah, Kolkata, and travel by pre-booked reserved second-class seats on the Chittaranjan line (about four hours), with group tickets arranged through IRCTC 60 days in advance. (The starting point may be altered to suit your location.)

Day 1: Rail journey to Chittaranjan; check in at a guest house; orientation on electric traction.
Day 2: Shop-floor visit: bogie assembly, traction motor winding, transformer shop, and final testing of electric locomotives.
Day 3: Visit to an automated signalling centre: electronic interlocking, centralised traffic control, track circuits, and the control panel room.
Day 4: Technical debrief, quiz, and return journey.

Category-wise expenditure (per student):
Rail fare (return): Rs. 1,000
Accommodation (3 nights): Rs. 2,400
Food: Rs. 1,800
Local transport: Rs. 600
Entry and safety gear: Rs. 400
Contingency: Rs. 300
Total: Rs. 6,500 (about Rs. 2,60,000 for 40 students)

Visitor safety rules: Wear helmets, safety shoes, and ear protection on the shop floor; stay behind yellow lines; never touch overhead lines, traction equipment, or tracks; do not cross rails; avoid photography in restricted areas; and move only with the guide.

Project Proposal: Ideation

AI-Based Resume Analyzer

Project Proposal: AI-Based Resume Analyzer

1. Technical Abstract

This project proposes an AI-powered Resume Analyzer System designed for students and the placement cell of the CSE department. The system automatically analyzes resumes (PDF and DOCX), extracts skills, education, projects, and experience using Natural Language Processing (NLP), and compares them with a target job description. Using text parsing, named entity recognition, TF-IDF and semantic similarity, and machine learning, the system calculates an ATS (Applicant Tracking System) compatibility score, identifies missing keywords and skill gaps, and gives section-wise suggestions for improvement. It generates a detailed PDF report that helps students fix their resumes quickly. Built with Python, the system runs on standard lab computers and aims to reduce resume review time for the placement cell by 60-70%.

2. Problem Definition and Context

The Problem:

Every placement season, hundreds of students in our CSE department prepare resumes, and faculty and the placement cell review them manually. This process is time-consuming, as each resume takes several minutes to read and give feedback on; inconsistent, as different reviewers have different opinions about what a good resume looks like; subjective, as students get vague comments such as "improve your resume" without knowing what exactly to fix; and hard to scale, as increasing student strength makes one-to-one review unsustainable. Many companies also use ATS software that rejects resumes automatically when keywords are missing or the format cannot be read, and students rarely know why they were rejected.

Real Impact:

Skilled students are rejected before any human reads their resume, students lose confidence after repeated rejections, the placement cell spends more time reviewing resumes than training students, and the college's placement results suffer because strong candidates are not presented well.

Why Existing Tools Don't Work:

Online resume checkers such as Jobscan, Resume Worded, and Zety usually offer only a few free scans and need a paid subscription for full feedback. They require internet access and the upload of personal data to third-party servers, which raises privacy concerns. Their advice is generic and not designed for fresher and CSE student resumes. The ATS systems used by companies are not available to students. Manual review is the obvious solution but is unsustainable.

Our Solution:

A lightweight, free tool built specifically for our CSE students that analyzes a resume against a job description, gives an ATS score, and generates clear, actionable suggestions. Because it

Project Proposal: Ideation

AI-Based Resume Analyzer

runs locally, student data stays private.

3. Engineering Methodology and System Core Logic

The system operates through a sequential pipeline. It begins with **text extraction**, where the uploaded resume is read using pdfminer and python-docx, with an OCR fallback for scanned resumes. **Preprocessing** then removes special characters and extra whitespace, converts text to lowercase, removes stop words, and applies lemmatization so that words like "developing" and "developed" match. **Section detection** splits the resume into Education, Skills, Projects, Experience, and Certifications using heading patterns.

Information extraction uses spaCy named entity recognition and regular expressions to pull out name, email, phone number, links, degrees, and organizations, while skills are matched against a curated skills dictionary covering programming languages, frameworks, databases, and tools. **Job description matching** compares the resume with the job description using TF-IDF cosine similarity for keyword overlap and sentence embeddings for semantic similarity, so that related terms such as "ML" and "machine learning" are understood. Missing keywords and skill gaps are listed.

ATS scoring combines weighted factors into a score from 0 to 100 percent: skills match, keyword coverage, project and experience relevance, education, section completeness, formatting, and use of action verbs and measurable achievements. **Machine learning enhancement** trains a classifier on a labeled dataset of strong, average, and weak resumes using features such as skill match score, number of projects, section completeness, resume length, and quantified achievements, so that the model predicts the overall resume quality. Finally, **report generation** shows the score breakdown, matched and missing keywords, section-wise suggestions, and exports everything as a PDF report.

4. Core Technical Stack

Programming Language: Python 3.9 selected for its extensive NLP libraries, strong machine learning ecosystem, and ease of development with rich community support.

Computer Vision Software Engine: Limited use. OpenCV and Tesseract OCR are used only as a fallback to read scanned or image-based resumes. The core of the project is text analysis rather than image processing.

Software Engineering: Flask web framework for backend API development, REST architecture for client-server communication, MVC pattern for clean separation of concerns, Git for version control, and Agile methodology for iterative development with weekly sprints.

NLP and Machine Learning: spaCy and NLTK for text processing, scikit-learn for TF-IDF and the classifier, and Sentence-Transformers for semantic similarity.

Microcontroller Infrastructure: Not required for this software-only project. The system runs on standard x86 computers available in college labs.

Hardware Switching Module Components: Standard Ethernet switches and WiFi routers for connectivity within the college LAN. No specialized switching hardware is required.

Project Proposal: Ideation

AI-Based Resume Analyzer

Data Transfer Middleware Channel: HTTP/HTTPS protocol for web-based resume uploads, RESTful API endpoints for data exchange between frontend and backend, JSON format for structured data transfer, and SQLite database connector for persistent storage of resumes, scores, and reports.

5. Presentation Delivery Script

[Slide 1: Title]

“Good morning respected guide, faculty, and friends. I'm [Your Name], presenting our final year project: AI-Based Resume Analyzer.”

[Slide 2: Problem]

“Every placement season, hundreds of students send resumes, but many are rejected before a human even reads them. ATS software filters out resumes with missing keywords or poor formatting, and students never get clear feedback. Meanwhile, our faculty cannot review every resume in detail.”

[Slide 3: Solution]

“Our system analyzes a resume automatically and compares it with a job description. Students upload their resume, and within seconds get an ATS score, a list of missing skills and keywords, and clear suggestions on what to improve.”

[Slide 4: How It Works]

“First, we extract text from the resume and clean it. Then we detect sections and extract skills, education, and projects using NLP. We compare the resume with the job description using TF-IDF and semantic similarity. Finally, a machine learning model rates the resume quality and the system generates a report.”

[Slide 5: Tech Stack]

“Built with Python, using spaCy for NLP, scikit-learn for machine learning, and Flask for the web interface. It supports PDF and DOCX resumes, runs on any lab computer, and keeps student data private.”

[Slide 6: Demo]

“Let me show you. [Demo: Upload a sample resume and a job description]. The system shows an ATS score, highlights the matched skills, lists the missing keywords, and suggests improvements for each section.”

[Slide 7: Impact]

“This helps students fix their resumes before applying, saves the placement cell review time, and gives every student the same fair and consistent feedback. [Add your tested results here.]”

[Slide 8: Thank You]

“Thank you. We welcome your feedback and questions.”

6. Strategic Evaluation

Why This Project is Unique:

Project Proposal: Ideation

AI-Based Resume Analyzer

This project is built for CSE students and understands technical resumes with projects, programming languages, and tools. It is free and works without a subscription. It protects privacy because resumes are processed locally. It is accurate, as it uses semantic matching and not only exact keyword counting. It provides actionable feedback showing exactly what to add or change, not just a score.

Challenges and Solutions:

- **Different resume formats** (columns, tables, templates) are handled by robust text extraction and an OCR fallback.
- **Different words for the same skill** are handled by a skill dictionary with synonyms and by semantic similarity.
- **Fresher resumes with little experience** are scored with more weight on projects, skills, and certifications.
- **Biased or unfair scoring** is reduced by using a transparent weighted formula, a trained model, and no personal details such as name or gender in the score.
- **Student adoption** is ensured through a simple web interface requiring minimal training.

7. Benchmarks and Institutional Impact

The system is designed to meet the following target benchmarks, which will be confirmed through testing on a dataset of sample resumes: resume quality classification accuracy of 85% or above, analysis of each resume in under 5 seconds on average, support for 2 file formats (PDF and DOCX), skill extraction accuracy of 90% or above, and a reduction in resume review time of 60-70%.

A pilot test with students from the CSE department is planned to measure how much their ATS scores improve after applying the suggestions. For the institution, the system gives students consistent feedback, lets the placement cell focus on training instead of repetitive review, and improves the quality of resumes presented to recruiters, which strengthens the department's placement outcomes.

LinkedIn and GitHub Profiles

Part A: LinkedIn profile

CONTACT

Email
irfairfa0810@gmail.com

LinkedIn
linkedin.com/in/irfana-s-a9205b43b

GitHub
github.com/irfana0810

TOP SKILLS

- C Programming (Beginner)
- Programming Fundamentals
- Logical Problem-Solving

AREAS OF INTEREST

- Machine Learning
- Generative AI
- Predictive Modeling
- AI Applications
- Chatbots

LANGUAGES

English
Tamil

IRFANA S.

Aspiring Machine Learning Engineer | Artificial Intelligence & Machine Learning | Generative AI

SUMMARY

I am focused on developing expertise in Machine Learning Engineering, Artificial Intelligence and Generative AI, with a strong interest in building intelligent applications that solve real-world problems. I am currently strengthening my programming fundamentals, logical thinking and problem-solving abilities while preparing to learn Python, data analysis, statistics and machine learning concepts.

CAREER OBJECTIVE

To become a skilled Machine Learning Engineer, focusing on predictive modeling, data-driven solutions and reliable AI systems, and to contribute to innovative AI/ML solutions that create real-world value.

EDUCATION

B.E. Computer Science and Engineering – St. Joseph's College of Engineering, Chennai (First Year, joined 2026). Higher Secondary Education – Bio-Maths.

CURRENTLY LEARNING

- C programming fundamentals
- Programming logic and problem-solving
- Preparing to learn Python
- Building a foundation for data analysis, statistics and machine learning
- Understanding Generative AI and AI application development

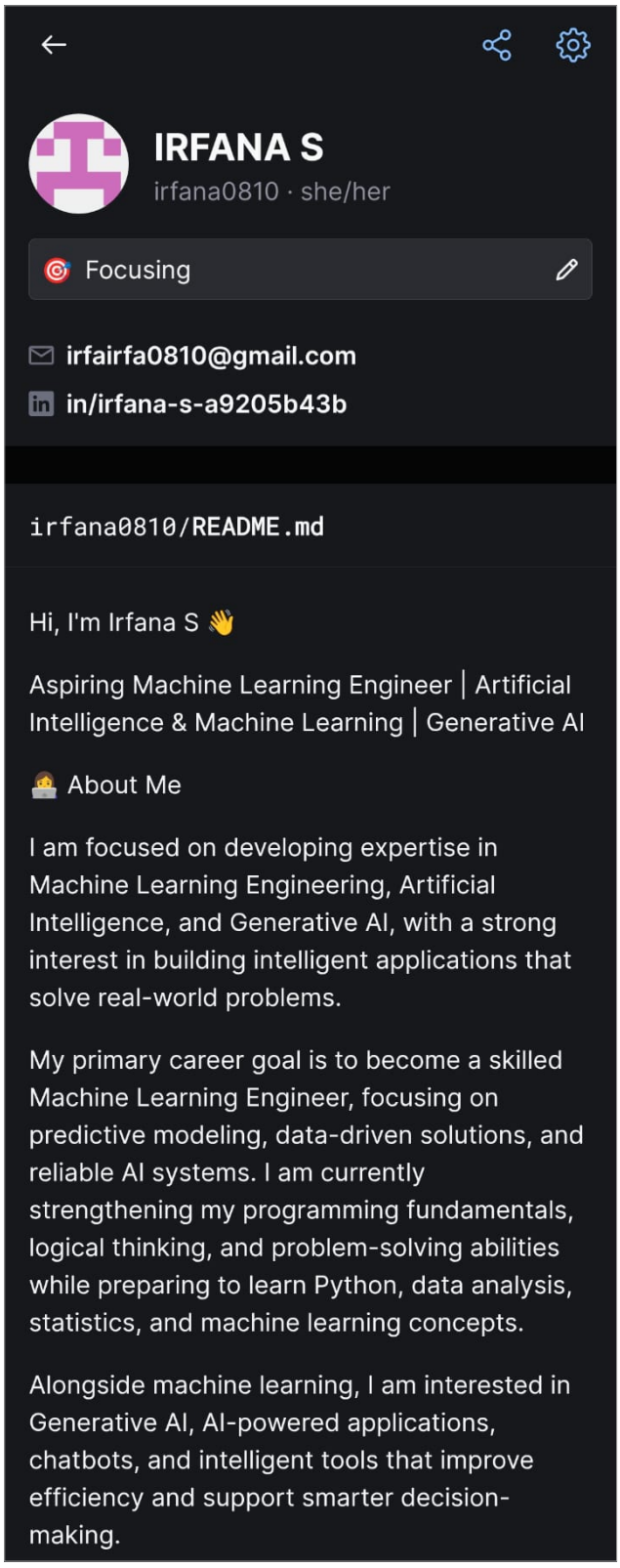
PROJECTS

No technical project completed yet. Planned: beginner-friendly projects such as a basic prediction system or an AI-powered study assistant. Project proposal prepared: AI-Based Resume Analyzer (Experiment No: II-7).

ONLINE PROFILES

LinkedIn: linkedin.com/in/irfana-s-a9205b43b
GitHub: github.com/irfana0810
Email: irfairfa0810@gmail.com

Part B: GitHub profile



IRFANA S

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🎯 Focusing

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irfana0810/README.md

Hi, I'm Irfana S 👋

Aspiring Machine Learning Engineer | Artificial Intelligence & Machine Learning | Generative AI

👩‍💻 About Me

I am focused on developing expertise in Machine Learning Engineering, Artificial Intelligence, and Generative AI, with a strong interest in building intelligent applications that solve real-world problems.

My primary career goal is to become a skilled Machine Learning Engineer, focusing on predictive modeling, data-driven solutions, and reliable AI systems. I am currently strengthening my programming fundamentals, logical thinking, and problem-solving abilities while preparing to learn Python, data analysis, statistics, and machine learning concepts.

Alongside machine learning, I am interested in Generative AI, AI-powered applications, chatbots, and intelligent tools that improve efficiency and support smarter decision-making.

My goal is to build practical technical skills, develop meaningful projects, and contribute to innovative AI/ML solutions that create real-world value.

🎯 Career Focus

- Primary Specialization: Machine Learning Engineering
- Additional Interest: Generative AI
- Areas of Interest: Predictive Modeling, AI Applications, Intelligent Systems, and Chatbots

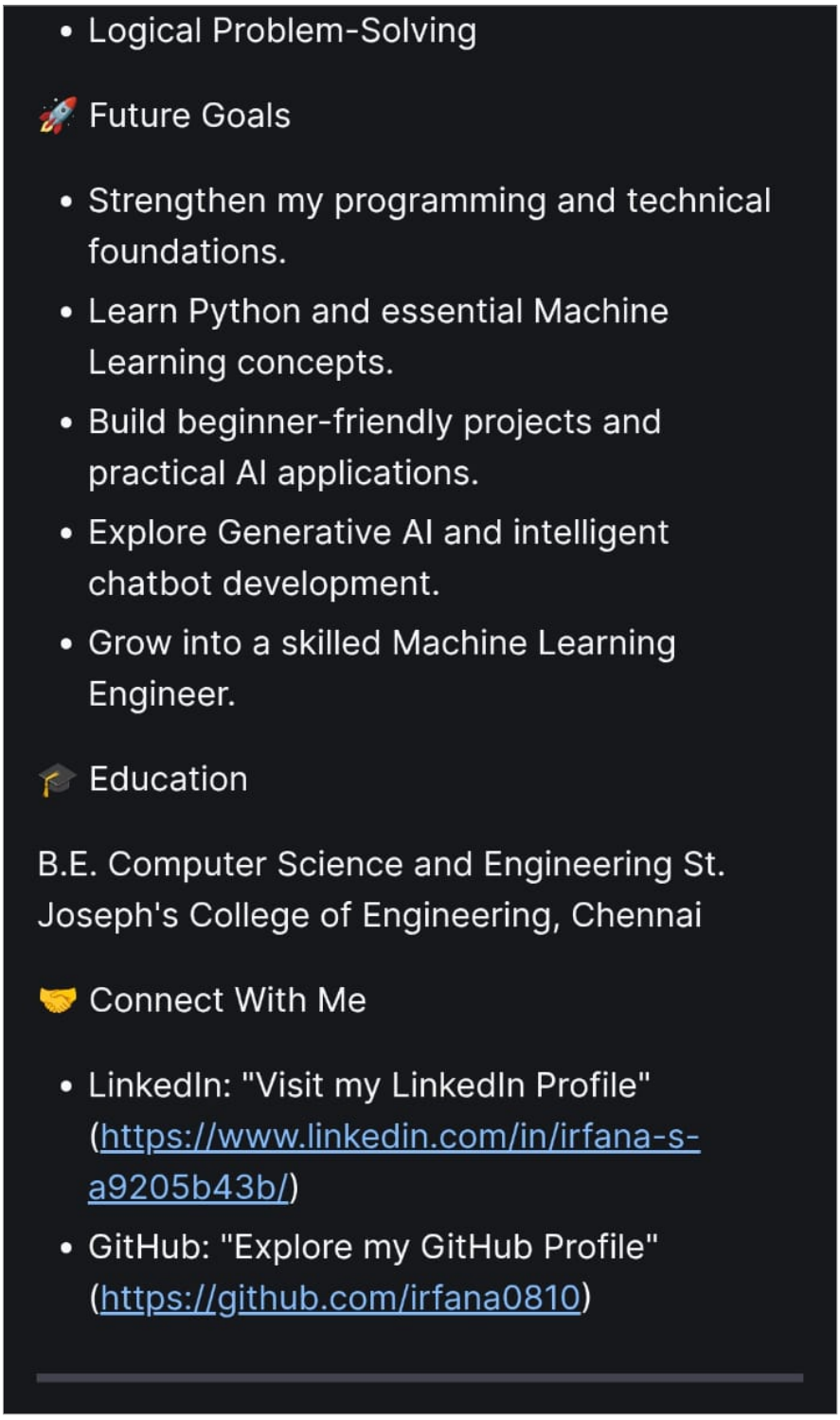
📚 Currently Learning

- C Programming fundamentals
- Programming logic and problem-solving
- Preparing to learn Python
- Building a foundation for data analysis, statistics, and Machine Learning
- Understanding Generative AI and AI application development

🛠️ Skills

- C Programming — Beginner
- Programming Fundamentals

Part B: GitHub profile



- Logical Problem-Solving

🚀 Future Goals

- Strengthen my programming and technical foundations.
- Learn Python and essential Machine Learning concepts.
- Build beginner-friendly projects and practical AI applications.
- Explore Generative AI and intelligent chatbot development.
- Grow into a skilled Machine Learning Engineer.

🎓 Education

B.E. Computer Science and Engineering St. Joseph's College of Engineering, Chennai

🤝 Connect With Me

- LinkedIn: "Visit my LinkedIn Profile" (https://www.linkedin.com/in/irfana-s-a9205b43b/)
- GitHub: "Explore my GitHub Profile" (https://github.com/irfana0810)